# 2 EXERCISE



## 4 EXERCISE PROTOCOLS

1. Walk as Much as Possible Each Day
2. Resistance Train Three Times per Week
3. Do Cardiovascular Training Three Times per Week
4. Incorporate Brief Mobility and Flexibility Work Two to Four Times per Week

Unofficial non-profit edition

I got really lucky with learning about exercise, and enjoying exercise—but only because I got unlucky first.

Like most kids in my town, I swam and played soccer, but unlike most I got really into skateboarding. I was decent enough at those other activities, but I loved skateboarding best. It felt the most liberating because I could do it any time of day or night, and I loved that there was no official court or playing field—city streets, curbs, benches, and stairs were as varied as they come. And it was transportation too, so it offered all sorts of possibilities for adventure. I could do it alone or with others, and I loved that you could innovate by making up new tricks. The sport was still relatively small at the time (this was the late 1980s and early 1990s) and, as I mentioned in the introduction to this book, several of my friends turned professional just out of high school. Not me. I kept getting hurt. By the time I was a sophomore, I had broken the same foot twice and messed up the arches of my feet.

And that's how exercise came into my life. Luckily, a supportive physical education teacher named Bob Peters noticed how often I arrived to class on crutches and pointed me toward a small weight room near the pool. No one really used it except the football team, and that was only during the summer months. So in many ways, it was a perfect situation. I could go in there and use the equipment whenever I wanted

and I didn't have to compare my (lack of) strength to anyone else because I was mostly alone. Although I didn't realize it at the time, the dearth of other people weight training also saved me from the terrible, often dangerous advice people tend to give others in gyms.

The only problem was that I had absolutely no idea what I was doing. There were anatomy charts of muscles and bones and tendons on the walls, and a few diagrams of exercises that were easy enough to follow, but nothing else. That drove me to the library and local bookstore, where I read about weight training. Which left me even more confused. The recommendations were all over the place. "Do one set per body part but super slow." "Do twenty sets per body part." "Train for thirty minutes maximum, three times a week." "Train for two hours, twice a day, six days per week," and on and on. It was dizzying. I completely empathize with people who experience immense confusion when they try to figure out how best to exercise, because I felt that too.

Luckily (again), one of the books I stumbled upon was by Arthur Jones, the inventor of Nautilus exercise machines. Jones was a proponent of doing short, intense workouts two to three times per week, or even less. While that didn't sound like enough to get results, I liked that he had a rationale for his suggestions and I liked that his general training

philosophy was based on science—in particular the work of physiologist Hans Selye, who studied stress and adaptations to stress.[1] (More about Selye in the next chapter…)

Exercise, Jones pointed out, is a form of stress that the body adjusts to with both specific adaptations, which reflect the type of stress that had been imposed to create them, and general adaptations, which come from pretty much any physical activity, to varying degrees.[2] Jones believed the goal of resistance training—or running, or any deliberate exercise for that matter—is the specific adaptations. The general adaptations that also occurred were a welcome bonus. Jones felt that when it came to resistance training, it didn't take much of the correct type of workout to stimulate the specific adaptations that would lead to, say, getting stronger or increasing the size of a muscle. And those two ideas formed the basis of his program.

I didn't know if Jones was wrong or right—after all, I had minimal experience with resistance training at that point. I also didn't have anyone to ask. Besides bodybuilders and football players—of which I was neither—not many people

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[1] Hans Selye, "Stress and the General Adaptation Syndrome," British Medical Journal 1, no. 4667 (June 1950): 1383–92, https://doi.org/10.1136/bmj.1.4667.1383.

[2] Nautilus Bulletin 1, ArthurJonesExercise.com, http://www.arthurjonesexercise.com/Bulletin1/37.PDF; Nautilus Bulletin 2, ArthurJonesExercise.com, http://www.arthurjonesexercise.com/Bulletin2/Bulletin2.html.

practiced resistance training back then, and most who did either looked big and puffy or veiny and really unnatural, neither of which was my goal. I wanted to get stronger and, at 6′1″ and about 150 pounds, I would not have minded having a bit more muscle too. The photos in Jones's books featured men and women who looked strong as well as lean. Some were younger, some were older. All that was a stark and welcome contrast to the extreme physiques highlighted in other weight training books and magazines.

And there was more: Jones emphasized that a good exercise program is one that doesn't take so much time that it disrupts one's school, work, and relationships. I liked the sound of that too. Hungry to learn more, I tracked down the phone number for one of Jones's protégés, Mike Mentzer, a legendary bodybuilder turned trainer from the late 1970s and early 1980s.[3] I called and left a message, and Mike called me back. I was sixteen at the time, and my mother was understandably worried. She wondered: Who was this grown man calling her teenage son at home? Apparently, Mike appreciated that when I called him, I rattled off a quick list of questions onto his answering machine that made it clear I had read Jones's books. We ended up speaking for several hours over the course of a few months about weight training,

[3] Kate Neudecker, "This Golden Era Bodybuilder Trained Just Twice a Week—and New Science Proves It Works," Men's Health, September 22, 2025, https://www.menshealth.com/uk/building-muscle/train-smarter/a67959687/mike-mentzer-heavy-duty-workout/.

fitness more generally, and science—both actual science and scientists. It wasn't as if he gently took me under his wing though. Mike was super intense. He would sort of bark at me when he spoke, but there was always a hint of kindness in his delivery. And he did charge me $150 for the calls, which was most of my savings at the time but it was well worth it.

I'm telling you this story because what I learned from those calls with Mike has stayed with me and I credit much of my regular progress with resistance training over the last thirty years to him. Here are ten counsels Mike gave me that I never forgot and always strive to implement with respect to resistance training.

- 1 After warming up, do not train for much more than sixty to seventy-five minutes at a time… you'll progress much faster that way. Occasionally your training may go longer if you are waiting for equipment at a gym, but that's just about the only reason to extend a session much beyond seventy-five minutes.
- 2 Always strive to use good form, meaning use a full range of motion for every repetition of every set, and control the weight from start to finish.
- 3 Learn to use resistance as a tool to stress the muscles you're trying to train. Focus as you do the lift to feel the muscles you're trying to train working. Mike added: If you can't feel the muscle you're trying to target you'll

want to use “pre-exhaustion” techniques when you train. I explain what pre-exhaustion is later, but as the name suggests, it involves doing an isolation exercise (where only one joint moves) immediately before you do a compound exercise (where multiple joints move) for the same muscle. Mike knew most people hate doing pre-exhaustion exercises because they require using less weight on the compound movement. But for getting results with otherwise hard-to-feel and slow-to-progress muscles, he was spot on.

- 4 If you’re new to resistance training, you’ll benefit greatly by taking one to three weeks to find the right movements that collectively allow you to target all the major muscles of your body and that work best for your body mechanics. Learn to execute those movements correctly using relatively light (for you) weights, meaning an amount of resistance where you can do ten to twelve repetitions, even though you could easily do five more. This one-to-three-week “break-in” period is about learning how to train before you start the real work. If you’ve been regularly training with weights for a while, I suggest you take a full week off. Stay out of the gym completely for seven, or even up to ten, days. You may feel like you’re standing still during this period but it’s critical to your long-term progress, and you’ll thank me later. Once you start your real workouts, it doesn’t take

many sets per muscle group to get the results you want, IF you do them with proper form… and intensity.

- 5 Intensity is defined as working with a weight that is heavy enough (for you) that it limits you to completing six to twelve repetitions maximum in good form. There are times to deliberately use heavier weights that limit you to three to five repetitions maximum, or lighter weights that permit you to do thirteen to twenty repetitions maximum. When and where to do that comes later. If you're not familiar with any of these terms, don't worry. I'll get into all this as the chapter unfolds. But what struck me looking back on these notes as I wrote this chapter is how many sets and exercises to do, and how often to train, was debated like crazy back then and it still is now. Mike was a bit extreme; he favored doing very few actual work sets of an exercise—meaning effortful sets, not warm-ups. In fact, he suggested doing even just one work set per exercise, and one to two exercises per muscle group. I will address the science-supported and practical reasons for doing a bit more than that later. But for now, let's take in Mike's points about a relatively low number of work sets, or what is referred to as a volume of sets. Because one thing most everyone agrees on now that Mike extolled is this: If done properly, you don't need a high volume of sets per exercise or muscle group, and that more is often

counterproductive. And so we continue, with more from Mike.

- 6 Train three times per week but know that your body gets stronger on the rest days in between training sessions, and mostly while you sleep. Hard training taxes your whole nervous system, too, not just the muscles and tendons and ligaments you trained, so to make the best progress, every twelve to sixteen weeks, or even every ten to twelve weeks, take one complete week off from resistance training to allow your muscles and nervous system to recover. You'll likely make much more progress in the long run.
- 7 Your body adapts to training. Making progress requires continually stressing your muscles by increasing the amount of weight you use for exercise over time. You can also do more repetitions with the same amount of weight, but once you can do ten to twelve repetitions with a given weight on a given exercise, using proper form, you need to increase the amount of weight in order to continue to make progress. That is because to grow in strength and size, a muscle must be stressed to its limit. This is known as progressive overload. To ensure you're doing this, take notes on the number of sets and repetitions you're doing for each workout, along with the weights you're using. This is the best way to make sure you're progressively overloading your muscles from one workout to the next.

- 8 Take each work set to failure, the point where you can't move the weight anymore in good form despite your absolute greatest effort. And sometimes you have to push past failure by getting assistance from a training partner who nudges the weight just a bit, or by immediately reducing the weight and doing two to three more repetitions until you hit failure again.
- 9 Learn to enjoy training hard. Don't socialize in the gym. Warm up, train, and get out! And undoubtedly the best and most classic Mentzer-ism of all:
- 10 "DO NOT LISTEN TO WHAT ANYONE ELSE TELLS YOU AS MOST PEOPLE WHO LIFT WEIGHTS ARE MORONS AND MOST OF WHAT THEY DO AND SAY IS TOTAL BS!"

(Mike literally yelled number 10 into the phone so loud that I winced and had to move the receiver away from my ear. Later I understood what words of wisdom and care those were. To be fair, there are many folks in the fitness education and training community who offer great advice, but Mike wasn't terribly off with this one.)

However coarse the delivery, those ten points were (and remain) gems that we can all benefit from. Again, as I said, I got lucky. We now have tons of peer-reviewed science to support the general logic of Mike's advice, although at the time, much of it ran countercurrent to the dominant views

about resistance training, most notably the popular idea at the time that “more volume is better.” As it turns out, quality reps and sets are what count, and it does not take much high-quality resistance training to trigger the specific adaptations Jones outlined in his book. (Note: Quality has many meanings, which we will cover repeatedly below.)

Of course, those ten points do not exhaustively cover all one needs to know about resistance training, and considering the results of studies that have been done since, there are some important caveats that I’ll address. What I can say, however, is that I took Mike’s general advice to heart.

After our first call, I took a full week off from any resistance training and then, even though I had been lifting for about six months, I devoted a few weeks to finding the free-weight, weight-machine, and body-weight-only exercises that worked best for me—meaning the ones I could do for full range of motion, pain free, and that allowed me to stress each of the major muscle groups. I found the right combination of compound (multi-joint) exercises and isolation (single-joint) exercises for each muscle group, like the shoulders, biceps, quadriceps, et cetera.

Then I started doing resistance training three times per week, training different muscles on each of those three days: calves and legs one day, then a day or two off, then chest, back, and abdominals (and neck), then a day off, then shoulders and

arms, another day off, and then back to calves and legs day. So again, three days total in the gym per week. I've done that body part–split pretty much consistently for thirty-five years now. Even if I am traveling, I find a gym to do these workouts. And if I go to a location where I can't find a gym, I'll do body-weight-only exercises, or I take the week off and stay active in other ways. While general activity is no substitute for resistance training, I have found that a week off approximately every four months can be beneficial. I can arrange it to happen during a vacation or other travel, or if I get sick—fortunately a rarity for me—I don't train. I rest up and stay out of the gym (and thus avoid getting others sick too!) and that becomes the week off. Approaching resistance training this way has allowed me to make consistent progress with about three hours per week of resistance training while still fully enjoying other aspects of life and avoiding injuries.

Before going any further I want to acknowledge something that Mike got somewhat wrong and that is important to highlight here. There is not one best way to resistance train! You won't find too many effective programs that recommend fewer than three days of resistance training per week, but there are some good ones that involve more days. Some programs, like the one described in this book, are based on a body part split in which you train different body parts on different days. And even those splits can vary. For instance some people like to train all their pushing muscles

of the upper body (chest, shoulders, triceps) on one day, and pulling muscles (back and biceps) on another day, and then legs a different day. Other programs have you train your whole body in every workout; others still alternate between upper and lower body workouts. All these can work. Below I explain the rationale for the split I use and suggest—because it offers some not-so-obvious benefits. The point stands, nonetheless, that there are a lot of ways to resistance train.

Yet there is a core set of principles upon which all people—women, men, inexperienced, experienced, young, old—must rely on to build an effective program. I define an "effective program" as one that people are likely to stick to over many years, and that works well for those also doing cardiovascular exercise, which we know is also vital to our health and provides distinct benefits from resistance training. Many people want to grow certain muscles and leave others as they are. Others want to grow all their muscles. An effective program should offer all those options too.

It also means a program that does not sap your energy but instead gives you more energy. After all, what good is a program that leaves you tired and hungry to an extent that inhibits your work and relationships? While it is normal (and reflective of an effective workout) to be tired after a training session, that fatigue should be somewhat transient, even that day. Your workout shouldn't leave you dragging. During

your work sets, you work hard—both mentally and physically—and you'll sweat and perhaps get a bit sore, but those are not the goal. Those are the by-products of quality workouts. You also want to still be able to think and walk (!) in the hours and days that follow.

I've come up with a template to help everyone. It's part of why I call it a foundational fitness program—so named because it's grounded in research, the ideas above and that follow below, and the core physiological principles that undergird all effective programs. These principles will never change because they reflect the mechanisms by which muscles, bones, and connective tissues get stronger and, if desired, grow larger. If you make sure that your resistance training follows this template, you can still tailor it to your goals, body type, schedule, and the phase of life you are in. That's what I mean by a foundational fitness program.

You might be thinking, "What about cardiovascular training?" There too I got really lucky. By the time I was seventeen, I was doing resistance training three times per week per Mike's instructions. I got great results in terms of strength and muscle size increases. And I enjoyed it. But I quickly found that I missed outdoor activities. Mike subscribed to the common idea back then that you couldn't make meaningful progress with resistance training while also doing endurance-type activities, like running or biking.

The reverse was also thought to be true. Nonetheless, I disobeyed Mike and what all other training books at the time said and I started running on the days that I didn't resistance train. I had an excellent reason to. My girlfriend lived about fifteen minutes away by car, I didn't have my own car yet, there was no bus to her house, and Uber didn't exist back then. So, much of the time I would just run to her house, which took me about forty-five to sixty minutes, thanks to all the hills. Sometimes I would bike there instead, which took about thirty minutes. Once again, I had no idea what I was doing. I would just set out and go. Ah, the joys of being young and naïve. And in love.

Here's where I got lucky: I was surprised at what happened. Instead of getting weaker, which is what everyone said would happen, I got stronger. And instead of resistance training making me slower at running and biking, which is what our school's cross-country coach told me would happen, I found I could run and bike faster and farther, and without any pain.

I also noticed that I always felt better mentally when I did a mix of cardiovascular and resistance training across the week, as opposed to just one or the other. I thought perhaps it was the fresh air… but I noticed this even if it was raining and I was forced to run on a treadmill.

I should mention that this was the early 1990s, and the ideologies within different sports and sectors of the fitness world were still very siloed. Today, we believe that being stronger helps in pretty much every sport and physical activity. Back then people also thought that if you did resistance training and then stopped, your muscles would turn to fat. Today we know that is false—in fact, it's impossible. Nowadays, most every serious athlete resistance trains during both the main and the offseason. And we now know that even if you don't play a sport, if your goal is to be healthy and fit, and to stay uninjured, and live as long as possible, and to do so with vigor, you're best off doing a combination of both resistance training and cardiovascular training. This is true for women and for men. We also know that what defines an effective resistance training program is the same for women and men, such as the need for intensity—training to failure on work sets (or close to failure)—as well as proper form, allowing sufficient rest between workouts for specific adaptations to occur, and more. As Lauren Colenso-Semple, a PhD in integrative physiology and a strength and conditioning coach, and one of the world's leading experts in strength and hypertrophy training, stated so clearly (and adamantly) when she was a guest on my podcast, all the data point to the fact that when it comes to resistance training to improve strength and/or muscle size, the same principles and techniques apply to

women and to men—including the need to train to failure or very close to it.[4] Looking back, for me it was thanks to those repeated foot injuries from skateboarding, the generosity of a physical education teacher, a few good books, some formative calls with one of the world's experts in resistance training, and being a naïve, lovestruck teenager that I learned early on how to get in and stay in great all-around shape. Lucky indeed.

And I promise you: If you follow the template here and add in cardiovascular training, too, you can get into and stay in great all-around shape at any age. Yes, nutrition matters but making exercise part of your life will absolutely make you a much healthier person, mentally and physically.

In this chapter I teach you how to do that. And if you already have a fitness program, I'm confident you'll still glean a ton of new knowledge that you can use from these pages.

Most people exercise because they want to feel and look good in their body. They also want longevity—not just to live as long as possible, but to be capable of doing any number of practical things without pain and, ideally, with

[4] Robert W. Morton et al., "Training for Strength and Hypertrophy: An Evidence-Based Approach," Current Opinion in Physiology 10 (2019): 90–95; Andrew Huberman, "Guest Series | Dr. Lauren Colenso-Semple: The Most Effective Weight Training, Cardio & Nutrition for Women," Huberman Lab (podcast audio), February 16, 2026, https://www.hubermanlab.com/episode/the-most-effective-weight-training-cardio-and-nutrition-for-women-lauren-colenso-semple.

relative ease. For example, having enough endurance for a long hike with a backpack. Being strong enough to lift boxes of books or help move furniture. Fast enough to sprint for (and catch) a plane or train without tearing a muscle. Able to open a jar when asked. The protocols here will enable you to do all that. They will also help you achieve the body composition (i.e., lean tissue-to-fat ratio) you want, while still allowing you to eat well, and feel more vital. Exercise is extremely potent medicine and, as with sleep, it's one of the best investments you can make toward your mental and physical health. The benefits start from the minute you begin each workout.

If all this gives the impression I love exercise… I do, on some days. Training is definitely a pleasure for me, especially if I've slept well, I'm hydrated, I'm caffeinated, and if there's no big stressor weighing on me. That's a lot of ifs, so it won't surprise you that those conditions are often hard to achieve because, well, life happens. I too sometimes get less-than-optimal sleep. I too have a full schedule. I too experience stress. I've had lapses due to the occasional cold. It happens to all of us.

That said, really understanding how exercise works and how much good can come from a few hours a week of focused training can keep us on the journey to get and stay fit when we are not feeling motivated. I look forward to workouts

because I know that the way I'm training works. That is priceless. Especially for people who don't like exercise, but like the feeling of having done it, knowing your time was well spent—that it will get you results—is a game changer. In fact, a really good program of the sort I describe here can allow you to make progress most every single week.

I mentioned earlier in this introduction that I empathize with people who experience immense confusion when they try to figure out how best to exercise, because I felt that too. So perhaps you're wondering: Why not just discuss "the best way to exercise" here? Maybe you're sensing a pattern: Just like there's no "best" way to resistance train, there's no best way to exercise. We know that resistance training is key. So is doing cardiovascular training. And we need mobility and some degree of flexibility too. There are literally thousands of different programs for each of these. But, if you understand the principles by which different types of exercise benefit us in different ways, you can come up with the truly best exercise program for you. As Andy Galpin, PhD, an expert in exercise physiology, shared with me on my podcast, "The methods are many, the principles are few."[5] Meaning there are many different ways to go about generating specific adaptations like strength and muscle size

[5] Andrew Huberman, "Guest Series | Dr. Andy Galpin: How to Assess & Improve All Aspects of Your Fitness," Huberman Lab (podcast audio), January 18, 2023, https://www.hubermanlab.com/episode/dr-andy-galpin-how-to-assess-improve-all-aspects-of-your-fitness.

increases, or improved heart stroke volume, which is the amount of blood your heart pumps. That's the power of understanding how the brain and body work, and that's why I explain the mechanisms behind each protocol.

I should say, too, that these principles are broad—I don't talk about specific demographics here. These are written to help the widest possible audience of women and men of any age. Know that I don't lump those two groups together casually. As mentioned above, the science shows that the principles of an effective exercise program are the same for men and for women. Of course, if ever the science points to key differences, I highlight them. But again, if you understand the mechanisms behind a protocol, you can personalize it for your specific needs. Likewise, if you are training for performance in a sport, you might adjust these, again, as you see fit.

## What is exercise?

Peering into the history of exercise turns out to be both interesting and informative. And helpful, when it comes to our health goals.[6] An Indian physician named Susruta, who lived around 600 BCE, was one of the first known practitioners to prescribe exercise. "It should be taken every

[6] Charles M. Tipton, "The History of 'Exercise Is Medicine' in Ancient Civilizations," Advances in Physiology Education 38, no. 2 (2014): 109–17, https://doi.org/10.1152/advan.00136.2013.

day," he wrote. "Diseases fly from the presence of a person habituated to regular physical exercise." Susruta's words are as relevant today as they were during his lifetime.

The problem is how to implement the prescription. Many people don't do it, or they do it wrong and end up discouraged, injured, or both. So many times I hear people say, "I used to work out every day," or "I used to be so fit," or "I don't have the time or energy to exercise anymore." What is rarely discussed is that exercise, done correctly, gives you energy. It extends your life (i.e., gives you more time). It also increases your level of focus for many hours afterward, which also saves you time. And there's more. Research shows that regular exercise can be as or even more effective for improving mood and lowering anxiety than prescription antidepressants or talk therapy.[7] On, and on, and

[7] Michael Noetel et al., "Effect of Exercise for Depression: Systematic Review and Network Meta-Analysis of Randomised Controlled Trials," BMJ 384, no. 1 (February 2024): e075847, https://doi.org/10.1136/bmj-2023-075847; James A. Blumenthal et al., "Exercise and Pharmacotherapy in the Treatment of Major Depressive Disorder," Psychosomatic Medicine 69 (September 2007): 587–96, https://doi.org/10.1097/PSY.0b013e318148c19a; J. A. Blumenthal et al., "Effects of Exercise Training on Older Patients with Major Depression," Archives of Internal Medicine 159, no. 19 (October 1999): 2349–56, https://doi.org/10.1001/archinte.159.19.2349; Gary Cooney, "Exercise for Depression," Journal of Evidence-Based Medicine 6, no. 4 (November 2013): 307–8, https://doi.org/10.1111/jebm.12076; Siri Kvam et al., "Exercise as a Treatment for Depression: A Meta-Analysis," Journal of Affective Disorders 202 (September 2016): 67–86, https://doi.org/10.1016/j.jad.2016.03.063; Ben Singh et al., "Effectiveness of Physical Activity Interventions for Improving Depression, Anxiety and Distress: An Overview of Systematic Reviews," British Journal of Sports Medicine 57, no. 18 (2023): 1203–9, https://doi.org/10.1136/bjsports-2022-106195.

on. Exercise is not a cure-all, but it is often a way to address many of the physical and mental health issues people struggle with. It's just that, as a species, we have not quite come to understand how potent a medicine exercise really is.

Then there are those who exercise consistently but overdo one type or just focus on their favorite sport, like running or tennis, Pilates or spin class, and miss out on the very real health benefits that come from variety.

In my experience, very few people know how to get in great all-around shape properly. Even fewer stay there. The good news is that doing so is pretty simple, again, once you know how.

It comes in three basic varieties: resistance training, cardiovascular exercise, and mobility and/or flexibility. Resistance exercise is where you use some weight—your body weight, weight machines, or free weights—to challenge your neuromuscular system and connective tissues. Resistance training also strengthens your bones. The hallmark of classic cardiovascular exercise is an increased heart rate for at least twelve minutes and up to seventy-five minutes—or in some cases even longer. There are arguments to be made that one can get effective cardiovascular workouts in shorter bouts of activity, through what's often called high-intensity interval training (HIIT). But we will return to that nuance later. Finally, mobility and flexibility

challenges our ability to stably shift between different bodily positions, and to have access to a full range of functional motion if we need it for athletics or simply in everyday life. If you're now thinking "So I have to do three types of exercise? I barely have time for one!" don't despair. Although it can be great to have mobility and flexibility practice, as I describe later in this chapter, you can see meaningful benefits from protocols that take very little time to complete, in one case just five minutes. And the resistance and cardiovascular sessions I describe below are aimed to give you the maximum benefits for time spent.

## What is the goal of exercise?

Believe it or not, the most important thing to understand about exercise is why you're doing it. I don't mean your psychology here. I am referring to what you are trying to achieve with each workout; why you are doing specifically what you are doing. It ensures every workout leads you to achieve your goals. For all that has been written and said about exercise, it often misses this crucial point. All effective forms of exercise share a common purpose: to trigger a set of specific and general adaptations in your body and brain.

Exercise is the stimulus, and the general and specific adaptations are your body's response to the stimulus. Exercise can also drive aesthetic changes, like fat loss, or improve sports performance, but those are secondary.

“Wait!” I hear you saying, “I exercise to stay fit. Just tell me how to do that.” I will, but I promise you will accomplish your goals much faster, and you will make consistent progress, if you understand exercise in this stimulus-adaptation context.

Here’s an example of how specific adaptations work. Let’s say you run or cycle at a moderate pace for thirty minutes. I’m talking about a pace where you can barely hold a conversation without gasping for air. While you do this your heart rate and blood pressure will rise substantially, you’ll breathe more rapidly, you’ll sweat, your muscles probably will burn a bit, and your immune system will release a bunch of pro-inflammatory molecules such as interleukin-6 into your blood.[8] Sounds terrible, right? Wrong. This constellation of responses—what we call the exercise-induced stress response—will lead to a set of adaptations that directly reflect the stress you just experienced but in the opposite direction. Thousands of coordinated, metabolic, molecular and even genetic changes occur when we exercise—a “molecular choreography,” as one group of researchers has put it.[9]

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[8] Bente Klarlund Pedersen and Anders Dyhr Toft, “Effects of Exercise on Lymphocytes and Cytokines,” British Journal of Sports Medicine 34 (2000): 246–51, https://doi.org/10.1136/bjsm.34.4.246.

[9] Jerzy A. Zoladz, Muscle and Exercise Physiology (Academic Press, 2018); Scott Powers and Edward Howley, Exercise Physiology: Theory and Application to Fitness and Performance, 10th ed. (McGraw Hill, 2017); MoTrPAC Study Group et al., “Temporal Dynamics of the Multi-Omic

After doing this for some time, your resting heart rate, blood pressure, and circulating levels of inflammatory molecules will be lower than they were before you set out on that run.

When most people think about what they accomplished by exercising, they tend to think about the calories burned or the mood enhancement it provides in the hours that follow. That's all well and good, but the real magic of exercise resides in the specific and general adaptations it produces. Those are the real "gains" of exercise. They last for days and they compound over time. They literally change your body and brain, and do so in ways that reflect the stress you experienced during the workout. That makes you a physiologically different person—capable of more in every area of life related to the exercise-induced adaptations—stronger, and faster, with more vigor and endurance, and more capable all around. Once you understand and experience the stimulus-adaptation process, you come to understand what effective effort is and when to

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Response to Endurance Exercise Training," Nature 629 (May 2024): 174–83, https://doi.org/10.1038/s41586-023-06877-w; Anna Maria Koman et al., "The Beneficial Effects of Aerobic Exercise on Human Systems and Organs: A Literature Review," Journal of Education, Health and Sport 73 (June 2024): 51710, https://doi.org/10.12775/JEHS.2024.73.51710; MoTrPAC Study Group et al., "Multi-Omic, Multi-Tissue Responses to Acute Exercise in Sedentary Adults: Findings from the Molecular Transducers of Physical Activity Consortium," bioRxiv, March 20, 2026, https://www.biorxiv.org/content/10.64898/2026.02.27.702183v2.full; Kévin Contrepois et al., "Molecular Choreography of Acute Exercise," Cell 181, no. 5 (May 2020): 1112–30.e16, https://doi.org/10.1016/j.cell.2020.04.043.

rest to glean the adaptations you worked to trigger. It can make you wiser too. In fact, as we will cover later in chapter 6—neuroplasticity, or your ability to learn and rewire your brain—follows the same general contour of "failures" (from genuine efforts to do something!) prompted by specific adaptations that will help you overcome those "failures" in the near future.

The effects of quality exercise on your ability to learn extend way beyond just metaphor. Exercise can directly facilitate neuroplasticity by allowing it to occur.[10] Like heat makes metal pliable, so too exercise brings the brain into a state where new neural connections in the brain and body can be forged.[11] Take for instance a 2011 study by a group of psychologists at the University of Pittsburgh. They randomized 120 older adults (average age sixty-seven) into two groups: one to cardiovascular exercise, the other to stretching only. The cardiovascular exercise group walked

[10] Silvia Grimanesa Revelo Herrera and Jose E. Leon-Rojas, "The Effect of Aerobic Exercise in Neuroplasticity, Learning, and Cognition: A Systematic Review," Cureus 16, no. 2 (2024): e54021, https://doi.org/10.7759/cureus.54021; Xueyang Li et al., "Physical Exercise for Brain Plasticity Promotion: An Overview of the Underlying Oscillatory Mechanism," Frontiers in Neuroscience 18 (August 2024): 1440975, https://doi.org/10.3389/fnins.2024.1440975; Lamia Ben Ezzdine et al., "Physical Activity and Neuroplasticity in Neurodegenerative Disorders: A Comprehensive Review of Exercise Interventions, Cognitive Training, and AI Applications," Frontiers in Neuroscience 19 (February 2025), https://doi.org/10.3389/fnins.2025.1502417.

[11] Kirk I. Erickson et al., "Exercise Training Increases Size of Hippocampus and Improves Memory," Proceedings of the National Academy of Sciences 108, no. 7 (2011): 3017–22, https://doi.org/10.1073/pnas.1015950108.

for increasing durations of time, reaching a maximum of forty minutes per session by week seven. By that point, the walking was brisk enough to get the participants' heart rates up to 50 to 60 percent of their maximum. And they did three sessions per week. The participants continued with that regimen for a year. According to the results, the hippocampus—a brain region associated with learning and memory—increased in volume by about 2 percent among the cardiovascular group, whereas it shrank by about 1.4 percent among the stretching-only group. While that might not seem like a lot, these findings are noteworthy because a shrinking hippocampus is linked to memory impairments during the later decades of life. A rather simple low-intensity exercise intervention seems to not only prevent brain deterioration but also trigger brain growth. Other studies, including those exploring higher-intensity exercise and resistance training show similar—and in some cases greater—neuroplasticity brought about by exercise.[12]

[12] K. I. Erickson et al., "Physical Activity Predicts Gray Matter Volume in Late Adulthood: The Cardiovascular Health Study," Neurology 75, no. 16 (2010): 1415–22, https://doi.org/10.1212/WNL.0b013e3181f88359; Tibor Hortobágyi et al., "Functional Relevance of Resistance Training-Induced Neuroplasticity in Health and Disease," Neuroscience and Biobehavioral Reviews 122 (March 2021): 79–91, https://doi.org/10.1016/j.neubiorev.2020.12.019; Fabian Herold et al., "Functional and/or Structural Brain Changes in Response to Resistance Exercises and Resistance Training Lead to Cognitive Improvements - A Systematic Review," European Review of Aging and Physical Activity 16, no. 10 (2019), https://doi.org/10.1186/s11556-019-0217-2; Orgesa Qipo et al., "Dose-Response Relationship of Resistance Training and the Effects on Circulating Biomarkers of Inflammation or Neuroplasticity In Older Adults: A

It perhaps shouldn't surprise us that exercise can open the gates for learning in the brain. A disproportionate amount of our brain's volume is devoted to generating our movements. Rodolfo Llinás, MD, PhD, a neuroscientist at NYU School of Medicine, cites an odd but telling example of the sea squirt, a spineless marine creature.[13] Young sea squirts swim freely until they are fully grown. The adults then stick themselves onto a rock and become filter feeders, catching whatever happens to float by. When this transition from swimming to sedentary occurs, sea squirts eat their own brains, leaving only what is necessary for their new passive style of feeding. Llinás's story about a distantly related animal is relevant to us for two reasons. First, it provides a warning to humans with an overly sedentary lifestyle. As humans age and become less mobile, their brains degenerate too, albeit not as dramatically as does the sea squirt brain.[14] Second, people who continue to exercise—in particular those who do exercise that is challenging (but safe) for them, offset much of typical age-related brain loss that is otherwise observed. This has real implications, allowing them to maintain youthful cognitive function well into their later

Systematic Review and Meta-Analysis," Ageing Research Reviews 113 (January 2026): 102917, https://doi.org/10.1016/j.arr.2025.102917.

[13] Rodolfo R. Llinás, I of the Vortex: From Neurons to Self (MIT Press, 2002), https://mitpress.mit.edu/9780262621632/i-of-the-vortex/.

[14] Emily Zhao et al., "The Role of Mobility as a Protective Factor of Cognitive Functioning in Aging Adults: A Review," Sports Health 6, no. 1 (2014): 63–69, https://doi.org/10.1177/1941738113477832.

years, possibly through neurogenesis, the addition of new brain neurons.[15] Deliberate, repetitive, effortful movements that elicit the short-term stress responses (discussed throughout this chapter) trigger a battery of chemical and neural feedback signals to the brain, which both supports the existing brain cells and the connections we would otherwise lose over time, as well as releases signals that cause neurons to grow new connections. This makes sense if you think about the sea squirt anecdote once more. Our brain is the most metabolically demanding organ in our body. Much of that organ is devoted to our ability to generate movement.[16] If we don't move, much of our brain is less active too. Am I telling you that we must exercise just to maintain the brain capacity we have? Indeed I am. Exercise—provided it is of sufficient intensity—makes us cognitively resilient.

The benefits of exercise to brain health and neuroplasticity turn out to be robust at every age, for women, for men, for people vulnerable to age-related cognitive decline, and for both neurologically healthy and unhealthy people alike.

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[15] Adria Quigley et al., "Effects of Exercise on Cognitive Performance in Older Adults: A Narrative Review of the Evidence, Possible Biological Mechanisms, and Recommendations for Exercise Prescription," Journal of Aging Research 2020 (May 14, 2020): 1407896, https://doi.org/10.1155/2020/1407896; Ahmed Disouky et al., "Human Hippocampal Neurogenesis in Adulthood, Ageing and Alzheimer's Disease," Nature 652 (2026): 1264–73, https://doi.org/10.1038/s41586-026-10169-4.

[16] Ziyue Aiden Wang et al., "Brain-Wide Analysis Reveals Movement Encoding Structured Across and Within Brain Areas," Nature Neuroscience 29 (2026): 147–58, https://doi.org/10.1038/s41593-025-02114-x.

Resistance training and cardiovascular training both impart huge benefits, but their combination (typically in separate workouts) turns out to be the most effective stimulus for brain health and neuroplasticity.[17] And when mobility and flexibility, and walking, are added in, the positive results increase significantly.

Some of the effects of exercise on brain health and neuroplasticity are the result of something called brain-derived neurotrophic factor, or BDNF, a protein our brain makes that can exert a powerful role on learning and memory by virtue of its ability to sculpt and reinforce new neural circuit connections.[18] Numerous studies have found that exercise leads to elevated levels of BDNF in the brain.[19] These elevations are temporary,[20] but enough to open the window for neuroplasticity for several hours after the exercise has ceased. Exercise also increases the release of

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[17] Li et al., "Physical Exercise for Brain Plasticity"; Ezzdine et al., "Physical Activity and Neuroplasticity."

[18] Magdalena Miranda et al., "Brain-Derived Neurotrophic Factor: A Key Molecule for Memory in the Healthy and the Pathological Brain," Frontiers in Cellular Neuroscience 13 (August 2019): 363, https://doi.org/10.3389/fncel.2019.00363; Carla Cunha et al., "A Simple Role for BDNF in Learning and Memory?," Frontiers in Molecular Neuroscience 3 (February 2010): 1, https://doi.org/10.3389/neuro.02.001.2010.

[19] Kristin L. Szuhany et al., "A Meta-Analytic Review of the Effects of Exercise on Brain-Derived Neurotrophic Factor," Journal of Psychiatric Research 60 (January 2015): 56–64, https://doi.org/10.1016/j.jpsychires.2014.10.003.

[20] Siresha Bathina and Undurti N. Das, "Brain-Derived Neurotrophic Factor and Its Clinical Implications," Archives of Medical Science 11, no. 6 (2015): 1164–78, https://doi.org/10.5114/aoms.2015.56342.

catecholamines:[21] that dopamine, epinephrine, norepinephrine trio that we discussed in the chapter on sleep and that we will revisit throughout this book. When that trio is deployed in the body and brain, our levels of alertness and focus are increased, allowing us to access the memories, skills, and cognitive abilities we harbor in our neural circuits. As neuromodulators, the catecholamines also can open the opportunity for neuroplasticity. (I explain this process in more detail in chapter 6.)

The way to think about the effects of exercise on brain health and neuroplasticity, therefore, is that moderate-to-intense exercise reinforces and maintains the brain functions we

[21] Hassane Zouhal et al., "Catecholamines and the Effects of Exercise, Training and Gender," Sports Medicine 38, no. 5 (2008): 401–23, https://doi.org/10.2165/00007256-200838050-00004; Joanna Kruk et al., "Physical Exercise and Catecholamines Response: Benefits and Health Risk: Possible Mechanisms," Free Radical Research 54, no. 2–3 (2020): 105–25, https://doi.org/10.1080/10715762.2020.1726343; William J. Kraemer and Nicholas A. Ratamess, "Hormonal Responses and Adaptations to Resistance Exercise and Training," Sports Medicine 35 (April 2005): 339–61, https://doi.org/10.2165/00007256-200535040-00004; Nikolaos Athanasiou et al., "Endocrine Responses of the Stress System to Different Types of Exercise," Reviews in Endocrine and Metabolic Disorders 24 (2023): 251–66, https://doi.org/10.1007/s11154-022-09758-1; David Walzik et al., "Molecular Insights of Exercise Therapy in Disease Prevention and Treatment," Signal Transduction and Targeted Therapy 9, no. 138 (May 2024), https://doi.org/10.1038/s41392-024-01841-0; R. S. Staron et al., "Skeletal Muscle Adaptations During Early Phase of Heavy-Resistance Training in Men and Women," Journal of Applied Physiology 76, no. 3 (1994): 1247–55, https://doi.org/10.1152/jappl.1994.76.3.1247; Vedat Çinar et al., "Impact of Different Exercise Modalities on Neuroendocrine Well-Being Markers Among University Students: A Study of Renalase and Catecholamine Responses," Frontiers in Physiology 16 (April 2025): 1591132, https://doi.org/10.3389/fphys.2025.1591132.

already have and lets us access the best of our attentional systems for focus and memory. It also opens the window for you to learn at an accelerated rate after the exercise session. Exercise alone won't make you smarter, but it opens the door for you to become smarter, should you decide to put in the effort to do so. (Chapter 6 explains the best and fastest way to do that.)

It's important to remember that any learning facilitated by elevated levels of BDNF is not confined to the neural circuits involved in the exercise that triggered it. Rather, BDNF will facilitate whatever learning we pursue while its levels are elevated. As one of many examples, just six minutes of intense running improved the ability of twenty-seven healthy adults to learn new words.[22] The improvements in learning and retention were tied to elevated levels of BDNF, along with elevated catecholamines: dopamine and epinephrine. Moderate-to-high-intensity exercise is arguably the best type of workout to trigger this process. People will debate whether resistance training is as effective at increasing neuroplasticity as cardiovascular training is. The key variable seems to be the intensity of training; it has to be sufficiently high. When done correctly, resistance training can be at least as taxing as intense

[22] Bernward Winter et al., "High Impact Running Improves Learning," Neurobiology of Learning and Memory 87, no. 4 (2007): 597–609, https://doi.org/10.1016/j.nlm.2006.11.003.

cardiovascular training when viewed through the lens of neurological demand, and perhaps more.[23]

Another way that exercise supports our brain and body is that it markedly improves sleep quality and duration.[24] Remember from chapter 1: Morning workouts appear to shorten the amount of time it takes to fall asleep at night, and regular exercise has been found to improve the overall quality of sleep for adults. In a study including forty-eight adults with insomnia, moderate aerobic exercise improved sleep, and helped the participants feel less stressed.[25]

One caveat. If exercise is very intense and done too close to your desired bedtime, it can be detrimental to sleep. (We cover this more in chapter 1 and chapter 3.) It's best to either avoid intense exercise in the hours prior to sleep, or if you can't, to make sure you engage in some deliberate downregulation practices, such as a few minutes of

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[23] Hortobágyi et al., "Functional Relevance"; Herold et al., "Functional and/or Structural Brain Changes."

[24] Jessica R. Alley et al., "Effects of Resistance Exercise Timing on Sleep Architecture and Nocturnal Blood Pressure," Journal of Strength and Conditioning Research 29, no. 5 (2015): 1378–85, https://doi.org/10.1519/JSC.0000000000000750; Giselle S. Passos et al., "Effect of Acute Physical Exercise on Patients with Chronic Primary Insomnia," Journal of Clinical Sleep Medicine 6, no. 3 (2010): 270–75, https://doi.org/10.5664/jcsm.27825; Majd A. Alnawwar et al., "The Effect of Physical Activity on Sleep Quality and Sleep Disorder: A Systematic Review," Cureus 15, no. 8 (2023): e43595, https://doi.org/10.7759/cureus.43595.

[25] Passos et al., "Effect of Acute Physical Exercise."

long-exhale breathing or a hot shower, so that it does not negatively impact your sleep.

There are many reasons why appropriately timed exercise improves sleep. As discussed in chapter 1, the chemical adenosine, which not only builds up in the brain and body throughout the day but is also further increased with exercise, creates an increasing pressure to sleep.[26] So exercise helps promote great sleep and reduces factors that lead to poor sleep. Regular physical activity also appears to increase the amount of time we spend in restorative slow-wave deep sleep, regardless of what time of day we exercise, and that translates to us feeling more rested when we wake.[27]

Then there is the benefit of moderate to intense exercise on our subjective waking experience, specifically on our feelings of well-being. Although "well-being" might sound nonscientific, for a lot of people it's an important barometer they use to gauge their quality of life. People who don't like exercise often look at me askance when I say that exercise

[26] Lynette L. Craft and Frank M. Perna, "The Benefits of Exercise for the Clinically Depressed," Primary Care Companion to the Journal of Clinical Psychiatry 6, no. 3 (2004): 104–11, https://doi.org/10.4088/pcc.v06n0301; Chaoxin Ji et al., "Physical Exercise Ameliorates Anxiety, Depression and Sleep Quality in College Students: Experimental Evidence from Exercise Intensity and Frequency," Behavioral Sciences 12, no. 3 (2022): 61, https://doi.org/10.3390/bs12030061.

[27] Insung Park et al., "Exercise Improves the Quality of Slow-Wave Sleep by Increasing Slow-Wave Stability," Scientific Reports 11, article no. 4410 (2021), https://doi.org/10.1038/s41598-021-83817-6.

improves mood, and not just because it helps us sleep better. But it's true: Exercise makes us happy.[28] And relaxed.[29] A moderate to intense workout is a proven, excellent way to snap ourselves out of a slump.[30] Exercise reduces our baseline levels of stress,[31] increases our capacity to show resilience in a wide range of circumstances,[32] and reduces anger.[33] These mood enhancements of exercise come from several different chemicals: dopamine, which increases motivation, focus, energy, and cognitive function;[34] and

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[28] Justy Reed and Deniz S. Ones, "The Effect of Acute Aerobic Exercise on Positive Activated Affect: A Meta-Analysis," Psychology of Sport and Exercise 7, no. 5 (2006): 477–514, https://doi.org/10.1016/j.psychsport.2005.11.003.

[29] Emma Childs and Harriet de Wit, "Regular Exercise Is Associated with Emotional Resilience to Acute Stress in Healthy Adults," Frontiers in Physiology 5 (May 2014): 161, https://doi.org/10.3389/fphys.2014.00161.

[30] R. E. Thayer et al., "Self-Regulation of Mood: Strategies for Changing a Bad Mood, Raising Energy, and Reducing Tension," Journal of Personality and Social Psychology 67, no. 5 (1994): 910–25, https://doi.org/10.1037/0022-3514.67.5.910.

[31] D. Scully et al., "Physical Exercise and Psychological Well Being: A Critical Review," British Journal of Sports Medicine 32, no. 2 (1998): 111–20, https://doi.org/10.1136/bjsm.32.2.111.

[32] Childs and de Wit, "Emotional Resilience to Acute Stress."

[33] Bonnie G. Berger and Robert W. Motl, "Exercise and Mood: A Selective Review and Synthesis of Research Employing the Profile of Mood States," Journal of Applied Sport Psychology 12, no. 1 (2000): 69–92, https://doi.org/10.1080/10413200008404214.

[34] Adilson Marques et al., "Bidirectional Association between Physical Activity and Dopamine Across Adulthood—a Systematic Review," Brain Sciences 11, no. 7 (2021): 829, https://doi.org/10.3390/brainsci11070829; Hamed Alizadeh Pahlavani, "Possible Role of Exercise Therapy on Depression: Effector Neurotransmitters as Key Players," Behavioural Brain Research 459 (February 29, 2024): 114791, https://doi.org/10.1016/j.bbr.2023.114791; Çinar, "Impact of Different Exercise Modalities."

serotonin, which does many things, including help make us more resistant to challenges.

And here's a fun fact: The lactate your body produces during high-intensity exercise can physically modify—or lactylate—proteins in the brain, altering how neurons communicate. A recent study found that lactate production spurred by exercise reduced anxiety-related behaviors in mice, apparently by supporting the molecular machinery present at synapses between neurons in the prefrontal cortex.[35] The prefrontal cortex has many parts and functions, but will come up repeatedly in this book in the context of reducing impulsivity, decreasing anxiety, and learning new information and skills.[36]

There's more. Exercise has a surprising effect on our energy levels. The relationship between exercise and energy is key to understand, especially for those who feel they don't have the energy to exercise. After all, exercise requires energy, right? Right. For those who don't enjoy it or who have a ton of demands on their schedule and may not be sleeping well or enough, that's especially true. Add in a bit of low-level or major depression, anxiety about finances or the state of the

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[35] Lan Yan et al., "Physical Exercise Mediates Cortical Synaptic Protein Lactylation to Improve Stress Resilience," Cell Metabolism 36, no. 9 (September 2024): 2104–17.e4, https://doi.org/10.1016/j.cmet.2024.07.018.

[36] Kathleen K. A. Cho et al., "Long-Range Inhibition Synchronizes and Updates Prefrontal Task Activity," Nature 617, no. 7961 (April 2023): 548–54, https://doi.org/10.1038/s41586-023-06012-9.

world, or challenges with relationships in your life and, voila: You have a person who quite understandably does not want to exercise, and very likely won't exercise, no matter how many studies about its benefits or inspirational tales they see and hear. If this is you or someone you know, I hope what I'm about to share will help.

Yes, exercise requires energy to initiate and to complete, but it also gives us energy. I'm not just referring to energy in the psychological sense. I mean it gives us actual chemical energy and it does so by changing the electrical state of your nervous system. Work from neuroscientist Peter Strick's laboratory, at the University of Pittsburgh, identified three sets of brain areas that, when activated, communicate with our adrenal glands to cause them to release adrenaline.[37] In turn, this adrenaline activates receptors on our vagus nerve, which is a dense neural pathway that bidirectionally connects our body and brain. Our vagus nerve then triggers the release of norepinephrine from the locus coeruleus, which in turn can increase release of dopamine, acetylcholine, and other neuromodulators in other brain centers.[38] Other research found that exercise enhances a

[37] Richard P. Dum et al., "The Mind-Body Problem: Circuits That Link the Cerebral Cortex to the Adrenal Medulla," Proceedings of the National Academy of Sciences 116, no. 51 (2019): 26321–28, https://doi.org/10.1073/pnas.1902297116.

[38] Teiko Miyashita and Cedric L. Williams, "Epinephrine Administration Increases Neural Impulses Propagated Along the Vagus Nerve: Role of Peripheral β-Adrenergic Receptors," Neurobiology of Learning and Memory 85,

group of neurons in the hypothalamus—known as VMH SF1 neurons—whose activity is required in order for endurance-specific adaptations to occur.[39] Meaning, the brain is what actually coordinates the specific bodily adaptations that effective exercise triggers. Put differently, your body changes because your brain tells it to.

While the abilities of our brain to energize and change our body are incredible, the key practical questions are: "What are the magical brain areas that, when made active, flick the

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no. 2 (March 2006): 116–24, https://doi.org/10.1016/j.nlm.2005.08.013; D. L. Hassert et al., "The Effects of Peripheral Vagal Nerve Stimulation at a Memory-Modulating Intensity on Norepinephrine Output in the Basolateral Amygdala," Behavioral Neuroscience 118, no. 1 (February 2004): 79–88, https://doi.org/10.1037/0735-7044.118.1.79; R. N. Holdefer and R. A. Jensen, "The Effects of Peripheral D-Amphetamine, 4-OH Amphetamine, and Epinephrine on Maintained Discharge in the Locus Coeruleus with Reference to the Modulation of Learning and Memory by These Substances," Brain Research 417, no. 1 (August 1987): 108–17, https://doi.org/10.1016/0006-8993(87)90184-3; Gina R. Poe et al., "Locus Coeruleus: A New Look at the Blue Spot," Nature Reviews Neuroscience 21 (September 2020): 644–59, https://doi.org/10.1038/s41583-020-0360-9; Mei Hao et al., "Neural Circuit Connections and Functions of Locus Coeruleus–Norepinephrine System," International Journal of Molecular Sciences 26, no. 22 (November 2025): 11163, https://doi.org/10.3390/ijms262211163; Avi K. Matarasso et al., "Stimulus-Dependent Dopamine Dynamics from Locus Coeruleus Axons," preprint available on bioRxiv (2025), https://doi.org/10.1101/2025.09.15.676390.

[39] Morgan Kindel et al., "Exercise-Induced Activation of Ventromedial Hypothalamic Steroidogenic Factor-1 Neurons Mediates Improvements in Endurance," Neuron 114, no. 9 (February 2026): 606–22.e17, https://doi.org/10.1016/j.neuron.2025.12.033; Teppei Fujikawa et al., "SF-1 Expression in the Hypothalamus Is Required for Beneficial Metabolic Effects of Exercise," eLife (November 2016): e18206, https://doi.org/10.7554/eLife.18206.

first domino in this brain–adrenals–vagus–locus coeruleus/basal ganglia circuit? And how do we activate those brain areas?" The answer, Strick found, is a set of brain regions that control voluntary movement of the major muscles of the limbs—and the core muscles; yes, your abdominals, or abs, but also the sheets of muscles below and all around them, such as the rectus abdominis and obliques. Translation: Deliberate movement of your limbs and your core muscles creates chemical energy that activates your brain to want to further move your body, and gives you heightened levels of motivation to continue doing so.

We can use that exercise-induced energy not just to exercise more or harder. It can be directed toward cognitive activities in the hours after you exercise or simply to go about the tasks of your day with more motivation and focus. Provided we don't exercise to the point of exhaustion, exercise markedly improves our brain function, psychological state, and overall energy levels. Yes, it involves hormones. Yes, it involves endorphins (everyone seems to know this based on the notion of a "runner's high"). But it also involves a shift in brain state.

And exercise affects more than just our brain. There's perhaps the most familiar benefit: our heart health. In 1986 researchers from Stanford and Harvard reported their findings from sixteen years of tracking physical activity and

the causes of death of nearly seventeen thousand people between the ages of thirty-five and seventy-four.[40] They found that the more a person walked, climbed stairs, or played sports, the less likely they were to die of heart disease. And the correlation was linear: The more physically active someone was, the lower the risk. This connection existed regardless of smoking status, weight, or having a parent who'd died young (a potential indicator of an inherited condition). The report, led by noted epidemiologist Ralph Paffenbarger, MD, and published in the New England Journal of Medicine, was considered a landmark—proof that exercise improved longevity, largely by preventing cardiovascular illnesses.

Research since confirms this finding. In one study, researchers followed 3,307 people with heart disease from 1985 to 2014.[41] Everyone in the study had been encouraged to exercise and lose weight. After thirty years, the researchers could see that while weight loss alone did not reduce the risk of death, exercise did. Other data confirm exercise's significance, even when people have other risk

[40] Ralph S. Paffenbarger et al., "Physical Activity, All-Cause Mortality, and Longevity of College Alumni," New England Journal of Medicine 314, no. 10 (1986): 605–13, https://doi.org/10.1056/NEJM198603063141003.

[41] Trine Moholdt et al., "Sustained Physical Activity, Not Weight Loss, Associated with Improved Survival in Coronary Heart Disease," Journal of the American College of Cardiology 71, no. 10 (2018): 1094–101, https://doi.org/10.1016/j.jacc.2018.01.011; Erratum, Moholdt et al., "Sustained Physical Activity," Journal of the American College of Cardiology 71, no. 13 (2018): 1499, https://doi.org/10.1016/j.jacc.2018.03.013.

factors. One group of researchers described this link starkly, back in 1996.[42] "Low fitness is an important precursor of mortality," they wrote. To put that in plain language: People who aren't active are more likely to die young.

One mechanism behind this link is increased insulin sensitivity.[43] Insulin is a hormone made and released from your pancreas that is essential for keeping your blood sugar, or glucose, levels within a particular range. Almost every cell in the body, including our neurons, requires glucose to function. Glucose comes from the food we eat and stored sources of energy, such as glycogen in our muscles and liver and adipose tissue, what is commonly referred to as body fat. Resistance to insulin leads to a buildup of glucose in our blood, which damages the endothelial cells that make up our capillaries, veins, and arteries, and deprives all cells of the glucose they need, impairing their functioning. It thus can

[42] Steven N. Blair et al., "Influences of Cardiorespiratory Fitness and Other Precursors on Cardiovascular Disease and All-Cause Mortality in Men and Women," JAMA 276, no. 3 (1996): 205–10, https://doi.org/10.1001/jama.1996.03540030039029.

[43] Stephen R. Bird and John A. Hawley, "Update on the Effects of Physical Activity on Insulin Sensitivity in Humans," BMJ Open Sport & Exercise Medicine 2 (March 1, 2017): e000143, https://doi.org/10.1136/bmjsem-2016-000143; Kedi Gao and Youliang Lin, "Insulin Resistance Mediates the Association Between Physical Activity and Mortality in US Adults with Metabolic Syndrome," Scientific Reports 15, article no. 18872 (2025), https://doi.org/10.1038/s41598-025-02921-z; Contrepois et al., "Molecular Choreography of Acute Exercise"; Lykke Sylow et al., "The Many Actions of Insulin in Skeletal Muscle, the Paramount Tissue Determining Glycemia," Cell Metabolism 33, no. 4 (April 2021): 758–80, https://doi.org/10.1016/j.cmet.2021.03.020.

cause cardiovascular disease and cerebrovascular diseases such as stroke and dementia. Exercise increases insulin sensitivity and can even improve the expression of genes that regulate insulin and glucose over long time spans which means more stable energy levels and reduced cellular inflammation and other symptoms of metabolic disorders.[44] People with diabetes are more sensitive to insulin after exercising,[45] too, and nondiabetics derive additional benefits from exercise-induced insulin sensitivity. Such benefits can be seen even after just one moderate workout.

What's more, exercise can make your muscles become much better at pulling glucose out of the bloodstream, which ultimately helps your body handle carbohydrates more efficiently.[46]

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[44] L. B. Borghouts and H. A. Keizer, "Exercise and Insulin Sensitivity: A Review," International Journal of Sports Medicine 21, no. 1 (2000): 1–12, https://doi.org/10.1055/s-2000-8847; Contrepois et al., "Molecular Choreography of Acute Exercise"; Sylow et al., "Many Actions of Insulin."

[45] Vighnesh Vetrivel Venkatasamy et al., "Effect of Physical Activity on Insulin Resistance, Inflammation and Oxidative Stress in Diabetes Mellitus," Journal of Clinical and Diagnostic Research 7, no. 8 (2013): 1764–66, https://doi.org/10.7860/JCDR/2013/6518.3306.

[46] Sylow et al., "Many Actions of Insulin"; Erik A. Richter and Mark Hargreaves, "Exercise, GLUT4, and Skeletal Muscle Glucose Uptake," Physiological Reviews 93, no. 3 (July 2013): 993–1017, https://doi.org/10.1152/physrev.00038.2012; Stephen M. Cornish and Matthew J. Barnes, "Editorial: Nutrition and Muscle Recovery After Exercise," Frontiers in Sports and Active Living 6 (April 2024): 1413822, https://doi.org/10.3389/fspor.2024.1413822; Lifeng Wang et al., "From Food Supplements to Functional Foods: Emerging Perspectives on Post-Exercise Recovery Nutrition," Nutrients 16, no. 23 (November 2024): 4081, https://doi.org/10.3390/nu16234081.

Finally (although I could go on!), there's this: As you now know from everything you've just read, exercise can make you healthier and happier, and it can allow you to get smarter too. It will also extend your life.

One analysis examined data from thirteen studies that directly compared life expectancy among active and inactive people.[47] Eight of those studies included both men and women, and the results showed an average increase of 3.7 years of life among people who exercised regularly compared to those who did not. While it's great to have a physically active job or lifestyle, the benefits of structured exercise are independent. You need both of these in your life.

And you don't need to have been a longtime fitness devotee either. A Swedish study of 2,205 middle-aged men found that people who started to exercise between the ages of fifty and sixty experienced the same life-extending benefit as men who'd been working out all their lives and continued to do so as they got older.[48] There are also now studies of people who started resistance training in their mid-sixties, seventies, and even eighties, and all made excellent progress in

[47] Carl-Detlef Reimers et al., "Does Physical Activity Increase Life Expectancy? A Review of the Literature," Journal of Aging Research 2012, no. 1 (2012): 243958, https://doi.org/10.1155/2012/243958.

[48] Liisa Byberg et al., "Total Mortality After Changes in Leisure Time Physical Activity in 50 Year Old Men: 35 Year Follow-Up of Population Based Cohort," BMJ 338 (March 5, 2009): b688, https://doi.org/10.1136/bmj.b688.

strength and derived many other general health and lifestyle benefits. In fact, the group in their eighties made the same degree of progress in muscle mass and strength compared to people who had been exercising regularly for decades![49]

There is a lot more to say about exercise and longevity, but the effects are multifaceted. Some of them arrive via improved metabolic health and fuel utilization. Some via improved sleep, and improved immune system functioning—again, assuming we don't chronically exhaust ourselves with over-exercising. The point is: Exercise gives us life—literally.

So by now I hope you're on board with the "why." We can move on to the "how."

## How should I exercise?

As I've mentioned, there is no one best way to exercise but there is, I believe, a best way to structure an effective exercise program. It starts with setting a foundation that you can build on—not necessarily by adding more sessions but by changing the type and amount of exercise, and the sequence in which you do them each week. Let me show you how.

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[49] Gabriel Nasri Marzuca-Nassr et al., "Muscle Mass and Strength Gains Following Resistance Exercise Training in Older Adults 65–75 Years and Older Adults Above 85 Years," International Journal of Sport Nutrition and Exercise Metabolism 34, no. 1 (October 2023): 11–19, https://doi.org/10.1123/ijsnem.2023-0087.

The data point to everyone—men and women, young and old—needing sessions specifically to maintain or build muscle and strength, as well as sessions to keep or improve cardiovascular fitness. Also, mobility and flexibility aren't just nice to have, they are crucial to ensure we can exercise for the rest of our lives relatively pain free, and just as importantly, to live full lives by virtue of the number and types of activities we can partake in. Sports, recreation, playing with our kids or grandkids, and even household chores require some degree of mobility and flexibility.

The five protocols that follow have everything you need to build an effective exercise program that can be customized for your age, fitness and aesthetic goals, and schedule. It makes the puzzle pieces I've outlined above fit together. They include the foundational fitness program I discussed: three days of resistance training—all relatively brief but intense workouts—and three days of cardio—each of different duration and intensity, and done on non-resistance training days. Plus daily walking that can be worked into your existing daily activities and some mobility and flexibility exercises that are fast, yet also effective. It is designed to be compatible with your life.

Yes, maintaining a consistent routine will require some commitment, even when a workout is the last thing you want to do. But remember: You will never feel worse for having

exercised. And completing a workout is much easier than you might think, if you keep in mind that all effective exercise is stress designed to create specific and general adaptations. Collectively, these are certain to make you more fit and healthy and capable than you ever imagined.




EXERCISE PROTOCOL

## Walk as Much as Possible Each Day

### WHAT TO DO

- Incorporate walking into your daily life. Also, in these instances in particular, if you can: Walk for five to ten minutes after meals, when health benefits of this activity are amplified.
- Walk or pace for at least five minutes for every thirty minutes of sitting or standing.
- Consider a walk-and-work desk. If you need to be at a desk for work for long stretches of time, a sit-stand desk may be a good investment. Even better is a walk-and-work desk, with a treadmill that fits below.
- Seven thousand steps per day is a good goal, and ten thousand is even better. Accelerometers, step counters, or even your phone can help you keep track, but are not necessary.

## HOW IT WORKS

This isn't that foundational fitness program I discussed; it's something even more essential. While walking isn't the only exercise we need to do, it's listed as the first protocol in this section because it should not be thought of as an add-on. Rather, it is at the bedrock of a healthy lifestyle, just as viewing morning sunlight is the basis of a good sleep routine. Simply walking for at least an hour a day (total, not even all at once) is one of the best things you can do to protect yourself against illness and improve your overall health. The data show us why.

A group of researchers from across the US recruited 2,110 people between the ages of thirty-eight and fifty years to wear an accelerometer, or a step counter, for one year.[50] The researchers then collected health data on these individuals for the next ten years (on average). Their analysis, published in JAMA Network Open in 2021, showed that people who took at least seven thousand steps per day during the decade of the study had a significantly lower mortality risk compared to people who took fewer steps.

In 2023, researchers from Kyoto University, in Japan, and UCLA published their study of 3,101 adults who took eight

[50] Amanda E. Paluch et al., "Steps per Day and All-Cause Mortality in Middle-Aged Adults in the Coronary Artery Risk Development in Young Adults Study," JAMA Network Open 4, no. 9 (2021): e2124516, https://doi.org/10.1001/jamanetworkopen.2021.24516.

thousand steps three to seven days per week, eight thousand steps one to two days per week, or fewer than eight thousand steps on any one day in their week.[51] Health data were collected over the ten years following study enrollment. Like the earlier report, this one found that taking eight thousand steps or more correlated with a significantly lower risk of death from all causes, including heart disease. The benefit appeared to plateau at three to four days per week of eight thousand steps.

And walking protects against more than heart disease. Researchers from Denmark and Australia investigated the link between walking and dementia risk.[52] They monitored 78,340 adults for an average of seven years and, like the studies above, saw a link between walking and a lower risk for dementia. They noted that the optimal "dose" of walking to reduce the risk of dementia was just shy of 10,000 steps per day. Walking has also been tied to reduced risks of

[51] Kosuke Inoue et al., "Association of Daily Step Patterns with Mortality in US Adults," JAMA Network Open 6, no. 3 (2023): e235174, https://doi.org/10.1001/jamanetworkopen.2023.5174.

[52] Borja del Pozo Cruz et al., "Association of Daily Step Count and Intensity with Incident Dementia in 78,430 Adults Living in the UK," JAMA Neurology 79, no. 10 (2022): 1059–63, https://doi.org/10.1001/jamaneurol.2022.2672.

depression,[53] diabetes,[54] cardiovascular disease,[55] and the incidence of some cancers,[56] as well as increased survival after certain cancer diagnoses.

A paper published in Nature Medicine in 2022 found that walking 8,200 steps or more per day had a preventive effect against obesity, sleep apnea, gastroesophageal reflux disease, and major depressive disorder.[57] That is a remarkable set of benefits for such a simple, zero-equipment, zero-cost activity. In other words: We should all be walking. A lot. If not every day, then almost every day.

A primary reason why walking is so beneficial is that it requires repeated muscular contractions. This activates an enzyme called AMPK. A 2018 paper by researchers at the

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[53] P. Bernard et al., "Effects of a Six-Month Walking Intervention on Depression in Inactive Post-Menopausal Women: A Randomized Controlled Trial," Aging & Mental Health 19, no. 6 (2015): 485–92, https://doi.org/10.1080/13607863.2014.948806.

[54] Marcel Ballin et al., "Daily Step Count and Incident Diabetes in Community-Dwelling 70-Year-Olds: A Prospective Cohort Study," BMC Public Health 20, article no. 1830 (2020), https://doi.org/10.1186/s12889-020-09929-2.

[55] Elaine M. Murtagh et al., "The Effect of Walking on Risk Factors for Cardiovascular Disease: An Updated Systematic Review and Meta-Analysis of Randomised Control Trials," Preventive Medicine 72 (March 2015): 34–43, https://doi.org/10.1016/j.ypmed.2014.12.041.

[56] Anne McTiernan et al., "Physical Activity in Cancer Prevention and Survival: A Systematic Review," Medicine & Science in Sports & Exercise 51, no. 6 (2019): 1252–61, https://doi.org/10.1249/MSS.0000000000001937.

[57] Hiral Master et al., "Association of Step Counts over Time with the Risk of Chronic Disease in the All of Us Research Program," Nature Medicine 28 (2022): 2301–8, https://doi.org/10.1038/s41591-022-02012-w.

Salk Institute for Biological Studies called AMPK the "guardian of metabolism," and with good reason.[58] This enzyme has multiple roles. It is involved in the process of stimulating glucose uptake into muscle cells. Animal research indicates that AMPK is also involved in generating more mitochondria through its interactions with PGC-1alpha, helps counter inflammation and improves vasodilation to help lower blood pressure and deliver more nutrients to cells, and under some conditions, can remove debris from them.[59] Taking enough steps each day is therefore beneficial for health in numerous ways that extend way beyond just the calories burned by walking.

Data also support the value of interrupting long bouts of sitting with short bouts of walking. A large review from 2022 gathered data from seven studies that examined the health benefits of interrupting long bouts of sitting with

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[58] Sébastien Herzig and Reuben J. Shaw, "AMPK: Guardian of Metabolism and Mitochondrial Homeostasis," Nature Reviews Molecular Cell Biology 19, no. 2 (2018): 121–35, https://doi.org/10.1038/nrm.2017.95.

[59] Sibylle Jäger et al., "AMP-Activated Protein Kinase (AMPK) Action in Skeletal Muscle via Direct Phosphorylation of PGC-1α," Proceedings of the National Academy of Sciences of the United States of America 104, no. 29 (July 2007): 12017–22, https://doi.org/10.1073/pnas.0705070104; Sandra Galic et al., "Hematopoietic AMPK β1 Reduces Mouse Adipose Tissue Macrophage Inflammation and Insulin Resistance in Obesity," Journal of Clinical Investigation 121, no. 12 (December 2011): 4903–15, https://doi.org/10.1172/JCI58577; Jianhui Li et al., "AMPK Alters Proteasome Phosphorylation Status and Prevents Persistent Proteasome Condensates," bioRxiv (July 2025), https://www.biorxiv.org/content/10.1101/2025.07.01.662652v1.

either standing or walking.[60] According to the analysis, standing between long stretches of sitting helps gradually lower glucose levels, but walking is far more effective.

Specifically, those researchers wanted to see which change is better at lowering glucose levels after meals. Walking is particularly effective right after a meal because glucose levels tend to increase at this time and preventing that rapid rise can help lower the risk of diabetes and heart disease.[61] (High glucose levels are linked to both.) A review of eight randomized studies examining thirty different interventions in 116 people found that walking within thirty minutes after a meal lowers glucose levels more effectively than walking before a meal or after a break.[62] That's because we use glucose as energy… to walk. Moving therefore helps clear glucose out of the bloodstream, which is both protective to our vascular cells—our arteries, veins, and capillaries—and

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[60] Aidan J. Buffey et al., "The Acute Effects of Interrupting Prolonged Sitting Time in Adults with Standing and Light-Intensity Walking on Biomarkers of Cardiometabolic Health in Adults: A Systematic Review and Meta-Analysis," Sports Medicine 52 (2022): 1765–87, https://doi.org/10.1007/s40279-022-01649-4.

[61] Tobias Engeroff et al., "After Dinner Rest a While, After Supper Walk a Mile? A Systematic Review with Meta-Analysis on the Acute Postprandial Glycemic Response to Exercise Before and After Meal Ingestion in Healthy Subjects and Patients with Impaired Glucose Tolerance," Sports Medicine 53 (2023): 849–69, https://doi.org/10.1007/s40279-022-01808-7; Yasuyo Hijikata and Seika Yamada, "Walking Just After a Meal Seems to Be More Effective for Weight Loss Than Waiting for One Hour to Walk After a Meal," International Journal of General Medicine 4 (June 2011): 447–50, https://doi.org/10.2147/IJGM.S18837.

[62] Engeroff et al., "After Dinner Rest a While."

is beneficial for all the cells of our body by making them more effective energy utilizing machines, as described above. There is no need to rest right after eating. Just head straight out for that walk, even for just five minutes, although ten to twenty or even thirty minutes would be better.

We are talking about walking, not a workout. Just low-level, repeated muscle contractions. In fact, these contractions don't even have to be created by walking: A 2022 study from the University of Houston found that repeated contractions of the soleus muscle, the deep muscle of our calf, significantly improve glucose regulation.[63] Remarkably, this was found to be the case even if people remained seated and repeatedly raised their heel off the floor. It seems that even though the soleus is only about 1 percent of our total musculature, it has an outsized (positive) effect in regulating our glucose levels when it is repeatedly contracted by walking or "soleus pushups." Of course, doing repeated soleus contractions while seated throughout the day, while possible, is not terribly compatible with daily life. Most people would not be able to concentrate on that while doing their work. But I include this recent finding here to emphasize just how powerful low-impact-repetitive muscle

[63] Marc T. Hamilton et al., "A Potent Physiological Method to Magnify and Sustain Soleus Oxidative Metabolism Improves Glucose and Lipid Regulation," iScience 25, no. 9 (2022): 104869, https://doi.org/10.1016/j.isci.2022.104869.

contractions can be. And of course, people who spend hours in meetings all day may want to consider doing calf raises while they're confined to a desk. It's not a replacement for walking, but it is certainly better than sitting still for long amounts of time.

What about standing still for long stretches of time? The study just described tells us that standing breaks are beneficial. Maybe you have a standing desk at your office, and that can be a great way to stretch your legs throughout the day. But walking is definitely better. A walk-and-work desk—where a treadmill fits underneath—can be a great way for some people to walk more.[64]

Curiosity about the health benefits of walking has generated a tremendous amount of data. As mentioned above, we now have studies showing that walking reduces the risk of heart disease, hypertension, type 2 diabetes, cancer, cognitive impairment, death by pneumonia, poor sleep, mental health issues, stress, hormonal imbalances tied to aging, and more.[65] Glucose regulation is certainly a part of it, but a deep dive into the molecular mechanisms behind all these effects is beyond the scope of this book. For those who are

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[64] James A. Levine and Jennifer M. Miller, "The Energy Expenditure of Using a 'Walk-and-Work' Desk for Office Workers with Obesity," British Journal of Sports Medicine 41 (2007): 558–61, https://doi.org/10.1136/bjsm.2006.032755.

[65] Zoltan Ungvari et al., "The Multifaceted Benefits of Walking for Healthy Aging: From Blue Zones to Molecular Mechanisms," GeroScience 45, no. 6 (2023): 3211–39, https://doi.org/10.1007/s11357-023-00873-8.

interested, a multidisciplinary team of researchers published a comprehensive review of the cellular mechanisms behind this wide range of effects.

There's one more useful but nonobvious benefit to walking that I'd like to mention here. Walking induces optic flow, the state of having the visual image of the world slide past us. We are usually unaware of what this does to our nervous system. When we ambulate—nerd-speak for "walk"—a stream of visual input flows past us, an effect called global motion.[66] Human and animal studies have found that forward ambulation causes reflexive lateral and repetitive (right-left-right…) eye movements that serve to stabilize the visual image on the eyes. Perhaps surprisingly, such eye movements significantly reduce neural activity in the amygdala, a region of our brain involved in threat detection.[67] In fact repetitive lateralized eye movements are the cornerstone of trauma therapies such as EMDR (eye movement desensitization and reprocessing), which pairs left-right-left-right repetitive eye movements with the recounting of challenging experiences to uncouple their

[66] Diederick C. Niehorster, "Optic Flow: A History," i-Perception 12, no. 6 (2021): 20416695211055766, https://doi.org/10.1177/20416695211055766.

[67] Heekyung Jung and Andrew D. Huberman, "An Unbiased View of Neural Networks: More Than Meets the Eye," Neuron 100, no. 5 (2018): 1019–21, https://doi.org/10.1016/j.neuron.2018.11.038.

memory from the anxiety response.[68] EMDR is done in a therapist's office but was developed by psychologist Francine Shapiro, PhD, after she noticed that recounting traumatic experiences was made less stressful when the recounting took place on a walk.[69] EMDR is still considered somewhat controversial in the therapy community. Some believe EMDR's benefits come from the hypnosis-like state it can cause—and clinical hypnosis has well-established effects on mental health and brain rewiring.[70] Others believe EMDR has its own direct effects on relief from trauma and depression.[71] We need more data to parse all that. For now,

[68] Francine Shapiro, "The Role of Eye Movement Desensitization and Reprocessing (EMDR) Therapy in Medicine: Addressing the Psychological and Physical Symptoms Stemming from Adverse Life Experiences," Permanente Journal 18, no. 1 (Winter 2014): 71–77, https://doi.org/10.7812/TPP/13-098.

[69] Francine Shapiro, "Efficacy of the Eye Movement Desensitization Procedure in Treatment of Traumatic Memories," Journal of Traumatic Stress 2, no. 2 (1989): 199–223, https://doi.org/10.1002/jts.2490020207.

[70] Marco Pagani et al., "Eye Movement Desensitization and Reprocessing and Slow Wave Sleep: A Putative Mechanism of Action," Frontiers in Psychology 8 (November 2017), https://doi.org/10.3389/fpsyg.2017.01935; Leonard S. Milling et al., "A Meta-Analysis of Hypnotic Interventions for Depression Symptoms: High Hopes for Hypnosis?," American Journal of Clinical Hypnosis 61, no. 3 (December 2018): 227–43, https://doi.org/10.1080/00029157.2018.1489777; Siobhan K. O'Toole et al., "A Meta-Analysis of Hypnotherapeutic Techniques in the Treatment of PTSD Symptoms," Journal of Traumatic Stress 29, no. 1 (February 2016): 97–100, https://doi.org/10.1002/jts.22077.

[71] Ji-Woo Seok and Joong Il Kim, "The Efficacy of Eye Movement Desensitization and Reprocessing Treatment for Depression: A Meta-Analysis and Meta-Regression of Randomized Controlled Trials," Journal of Clinical Medicine 13, no. 18 (September 23, 2024): 5633, https://doi.org/10.3390/jcm13185633; Bessel A. van der Kolk et al., "A Randomized Clinical Trial of Eye Movement Desensitization and Reprocessing (EMDR), Fluoxetine, and Pill Placebo in the Treatment of Posttraumatic Stress

however, it is interesting to note that walking makes the amygdala less active, thereby reducing stress and anxiety.[72]

Alas, this benefit of walking won't happen if you're staring at your phone while doing it. That prevents the forward-ambulation-induced eye movements discussed here. The suppression of amygdala activity only occurs if we're looking forward and around. I believe this phenomenon is one reason why spending time simply walking or hiking in nature is so tied to well-being.[73] And also why walking or running on a treadmill—while better than not exercising—is less stress reducing than doing the same activity outdoors. The link between optic flow and stress reduction is an understudied area. However, walking outdoors is a zero-risk, zero-cost intervention that seems to translate to powerful results for your body and mental state.

## In My Experience

I have a step tracker on both my phone and my WHOOP band (I'm an advisor for WHOOP). I track my steps every

Disorder: Treatment Effects and Long-Term Maintenance," Journal of Clinical Psychiatry 68, no. 1 (January 2007): 37–46, https://doi.org/10.4088/jcp.v68n0105.

[72] Sonja Sudimac et al., "How Nature Nurtures: Amygdala Activity Decreases as the Result of a One-Hour Walk in Nature," Molecular Psychiatry 27, no. 11 (2022): 4446–52, https://doi.org/10.1038/s41380-022-01720-6.

[73] Mathew P. White et al., "Spending at Least 120 Minutes a Week in Nature Is Associated with Good Health and Wellbeing," Scientific Reports 9, article no. 7730 (June 2019), https://doi.org/10.1038/s41598-019-44097-3.

day but I don’t look at the data every day. I check it every few days to make sure I’m “hitting my numbers.” I know I get about seven thousand to ten thousand steps on average. More importantly, I am always seeking ways to take more steps. I make a point to pace indoors or walk outdoors whenever I take a phone call that doesn’t require me to be in front of my computer. I take stairs rather than escalators or elevators whenever I can. I see stairs as an opportunity, not a task, especially on travel days when I am going to be sitting a lot on the plane. I always strive to take two short walks of twenty to thirty minutes each during the day, usually one in the morning and one in the evening. If I miss the first one, I make the evening walk longer. I often take work calls, check in with friends or family, or listen to books or podcasts while I do. Every time we finish recording a podcast with a guest, our whole team takes a walk together (in the direction of the sun of course). If I have to sit a lot on a given day, I make it a point to walk a lot more the next day. I have an indoor treadmill for days when I can’t get outside to walk much due to work or weather.

I almost always use my standing desk to work at home. When I travel, I simply put my laptop on a box or stack of books on the desk or coffee table in the hotel room. It’s low-tech but it works, and I can rotate it toward the window to get more natural light. You need to find what works for you, of course.

## COMMON QUESTIONS

### How fast should I walk to get the health benefits of this activity

Any walking is better than no walking and brisk walking is better than slow walking.[74] Taking all studies discussed above into account, walking briskly for at least thirty minutes per day and ideally forty to sixty minutes should be the goal. Walking at least five days a week and making an effort to take stairs, et cetera, appears to be sufficient activity to meet current recommended guidelines.[75]

Again, walking is not "working out," but it is the basis of any good exercise program. Think of it as necessary for your health. We were made to move. Getting steps each day is essential and should be part of your life. You can weave it into time in the yard, time with friends, time on the phone, and so on. And when you are stuck in a meeting or on a plane and can't walk, try the calf raises discussed above. It's discreet enough that you're unlikely to get any strange looks.

Seven thousand steps are roughly an hour of walking, as I recommend, though obviously this will differ somewhat depending on the length of a person's stride, the terrain, and

[74] Xiaorong Bai et al., "Effect of Brisk Walking on Health-Related Physical Fitness Balance and Life Satisfaction Among the Elderly: A Systematic Review," Frontiers in Public Health 9 (January 31, 2022): 829367, https://doi.org/10.3389/fpubh.2021.829367.

[75] Ungvari et al., "Walking for Healthy Aging."

walking speed. And, again, these steps do not need to be done all at once but rather can be spread throughout the day.

Aim to speed up as you incorporate walking more into your life. If you already walk often, then try to walk a little faster. Faster walking obtains health benefits in less time.

### Are other types of low-impact movement, such as tai chi or slow swimming, as beneficial as walking

Yes and no. All movement is good movement, provided you don't get injured. Doing ten air squats—going from standing to a seated position with your thighs parallel to the ground, then standing up again, and repeating that ten times—every forty-five minutes throughout the day for an 8.5-hour period has been shown to have even better glucose-lowering effects than thirty minutes of walking, which makes it a decent alternative when you can't walk, like on really stormy days or on a long (or even a short) flight.[76] The ten-air-squats-every-forty-five-minutes approach obviously does not provide all the benefits that walking outside does (like optic flow, fresh air, sunlight, etc.).

There are other types of low-impact, continuous physical activity to explore. Tai chi is great for balance and is often done near sunrise and outdoors, adding to the benefit for

[76] Ying Gao et al., "Enhanced Muscle Activity During Interrupted Sitting Improves Glycemic Control in Overweight and Obese Men," Scandinavian Journal of Medicine and Science in Sports 34, no. 4 (April 2024): e14628, https://doi.org/10.1111/sms.14628.

reasons discussed in both chapter 1 and chapter 5, and it's typically done in groups, which has a positive social component. And tai chi certainly improves lower leg strength, balance, and coordination.[77] That's just one idea, but of course there are many others.

Everyone needs to find a low-impact, continuous physical activity that works for them in terms of enjoyment, and compatibility with lifestyle.

That being said, I'll come back to walking. Walking is a movement that draws on many different aspects of our anatomy: It puts weight on our skeletal structure, which is good for helping maintain our bone density and strength; it engages the soleus, which as mentioned before has a positive effect on glucose regulation; and it has the added bonus of optic flow for mental health. Add walking into your life wherever and whenever possible. Again, don't think of it as exercise, think of it as nonnegotiable movement that is foundational to a healthy day. Does that mean walking is the only thing we need to do? Of course not. In the context of physical activity, and to use scientific jargon, it is "necessary but not sufficient" for your health. So let's now discuss the

[77] Patricia Huston and Bruce McFarlane, "Health Benefits of Tai Chi: What Is the Evidence?," Canadian Family Physician 62, no. 11 (2016): 881–90, https://pubmed.ncbi.nlm.nih.gov/28661865/; Guo-Yan Yang et al., "Determining the Safety and Effectiveness of Tai Chi: A Critical Overview of 210 Systematic Reviews of Controlled Clinical Trials," Systematic Reviews 11, article no. 260 (December 3, 2022), https://doi.org/10.1186/s13643-022-02100-5.

things to do in addition to walking, starting with three times per week resistance training. Done properly.




EXERCISE PROTOCOL

## Resistance Train Three Times per Week

### WHAT TO DO

- Adhere to a program, starting with this foundational template that features three resistance training sessions every seven days.
- Day 1: Cardiovascular training (explained in Exercise Protocol 3)
- Day 2: Resistance training of major lower body muscles: quadriceps, hamstrings, calves Your glutes, core, and lower back are indirectly trained this day given the nature of the exercises involved, but direct glute training can be added.
- Day 3: Rest day
- Day 4: Resistance training of major torso muscles: chest, back, abdominals, and core (optional neck)
- Day 5: Cardiovascular training (explained in Exercise Protocol 3)
- Day 6: Cardiovascular training (explained in Exercise Protocol 3)

- Day 7: Resistance training of shoulders, biceps, triceps, abdominals/core (optional forearms and neck)

## HOW IT WORKS

This template is, in my experience, the best place to start and in many cases stay to achieve all-around fitness: strength, endurance, speed, and power. And it can be tailored to deliver the aesthetic changes you want. It is grounded in the fundamental principles of effective resistance training,[78] and I strongly believe it offers the best time-and-effort to results ratio. It can also be easily modified when your schedule gets busy or you need a week off. I have seen it work for everyone I've suggested it to.

This program consists of three sessions of resistance training every seven days. It also includes three cardiovascular training sessions, done on non–resistance training days. (These are detailed in the next protocol. I include them in the

[78] William J. Kraemer and Nicholas A. Ratamess, "Fundamentals of Resistance Training: Progression and Exercise Prescription," Medicine & Science in Sports & Exercise 36, no. 4 (2004): 674–88, https://doi.org/10.1249/01.mss.0000121945.36635.61; Brad J. Schoenfeld et al., "Effects of Resistance Training Frequency on Measures of Muscle Hypertrophy: A Systematic Review and Meta-Analysis," Sports Medicine 46, no. 11 (April 2016): 1689–97, https://doi.org/10.1007/s40279-016-0543-8; Eneko Baz-Valle et al., "A Systematic Review of the Effects of Different Resistance Training Volumes on Muscle Hypertrophy," Journal of Human Kinetics 81 (February 2022): 199–210, https://doi.org/10.2478/hukin-2022-0017.

template overview for clarity.) Together, these sessions provide the right amount of directed stress and recovery.

If you are already familiar with resistance training, you may think you need more sets to gain muscle or more training days for certain target areas than what this protocol suggests. Keep in mind that each session works in the larger context of the program. They synergize. For example, you'll train your legs but four days later, you'll sprint or row or cycle at an intensity that also taxes your legs (just enough, not too much). You won't lose ground.

For those new to resistance training, I highly suggest you do the "break-in" period, like the one detailed in Mike Mentzer's point number four in the introduction to this chapter. This will help you develop proper form with the exercises you select. For those experienced with resistance training, or novices who have completed the "break-in" period, you should design the specifics of each resistance training session with these variables as guidelines:

## WORKOUT DURATION: ABOUT 60 +/– 20 MINUTES

- Brief whole-body warm-up: About 5 minutes
- Total number of warm-up sets: 1–3 per exercise
- Total number of work sets per muscle group: 4–9
- Total number of exercises per muscle group: 2–3

- For each major muscle group: Do at least one exercise that is a compound (multi-joint) exercise and one that is an isolation (one-joint) exercise. A third compound movement can be added, if needed.
- Total number of work sets per exercise: 2–3 sets (usually just 2)
- Repetition (“reps”) range per set: As a general guideline, aim to use enough resistance so you reach “failure” after doing 6–10 reps of a given compound exercise, or in the case of an isolation exercise, after 8–15 reps. But if you can only get 5 reps, do that. Or if you can continue past 10 or 15 reps (reps in good form, that is!), keep going until you reach failure. Don’t stop simply because you’ve reached a given number of reps. Remember, too, that you must use progressive overload to get results. Once you can perform 10 reps of a compound exercise, or 10–15 of an isolation exercise, in perfect form, you’ll want to increase the amount of weight for work sets of that exercise in the next workout, which will very likely reduce the number of reps you can do until you progress further, and so on. This is absolutely key!

Note: The rep ranges discussed here and throughout this protocol are just guidelines; venturing into heavier 3–5 reps or lighter 15–25 repetition ranges makes good sense for

some exercises. So does systematically varying between very specific rep ranges for four to eight weeks at a time.

- Rest duration between warm-up sets: 1–2 minutes
- Rest duration between work sets: 2–5 minutes, depending on repetition range. Typically people rest longer (3–5 minutes) between work sets of compound exercises than between work sets of isolation exercises (90 seconds–3 minutes), and I suggest you do the same.

On reading the above list I imagine novices might already be confused and experienced resistance trainers already might be furrowing their brows, but let's consider each of these points so the rationale is clear.

## WORKOUT DURATION: ABOUT 60 +/– 20 MINUTES

You're in the gym to resistance train, and you're resistance training to trigger specific neural, muscular, and connective tissue adaptations. For most people the specific adaptations they seek are increased strength and perhaps also hypertrophy, or increased muscle size. Many want to also lose body fat.

It may surprise you to learn that body fat loss isn't a specific adaptation from doing a certain type of training. Why not? Because body fat loss is mostly the consequence of being in a caloric deficit. In chapter 4, I explain how certain foods

and food combinations often make us want to eat more or less, but it's caloric intake that determines if we maintain, lose, or gain weight.

Some people say they want to get toned, and others say they want to get jacked, or ripped. For the record: Toned is not an actual exercise adaptation either, but instead refers to a defined-yet-still-slender aesthetic. And while we're talking nonscientific but commonly heard fitness terms: Jacked means larger, rounded muscles and relatively lean, and ripped means still muscular and extra lean. Leanness comes from a combination of training and nutrition, but mostly nutrition. The program here is great for gaining muscle, maintaining muscle while losing body fat, or for gaining muscle and losing body fat, depending on how you adjust your caloric intake.

Many people of course also resistance train for the general adaptation benefits discussed earlier in this chapter, including improved brain and body health and longevity, elevated mood, better sleep, and so on.

Unless you're socializing or texting or taking endless selfies, or the gym is packed, your resistance training sessions should rarely last longer than about 60 +/– 20 minutes. After a five-minute whole-body warm-up, and then two to three warm-up sets for your first exercise, sixty to eighty minutes of effortful, highly focused "work sets" is more than enough

to stimulate the specific and general adaptations you're seeking. And that includes the rest periods between work sets too.

Why not longer? Because after eighty minutes of hard resistance training, the fatigue and stress on your body and nervous system is typically greater than is conducive for recovery before your next workout. Also, after eighty minutes, it can be hard to generate the focus and effort needed to continue to stimulate specific adaptations safely. While you might have the energy to train longer, it's good to finish your workout with some vigor left to devote to other aspects of life. Most people will find they won't progress much faster by training for longer sessions, or even more frequent sessions (and in many cases, they stop getting results altogether or even start regressing!). Especially when trying to incorporate the necessary cardiovascular training session discussed in Exercise Protocol 3. By resistance training three times a week for an hour or so, you're far more likely to be consistent over the long-term, which is ultimately what matters to both results now and in the future. There are times when because of long rest periods needed between sets in a crowded gym, workouts can take ninety minutes. Don't fret about it, but don't make that the norm. In fact, if you can get a great workout done in fewer than sixty minutes, even better.

Here are a few more nonobvious things that make a big difference to increase your efficiency in the gym:

## A note about “intensity” and soreness

People sometimes think that being really sore in the days after a workout is the goal, but it’s not. Delayed onset muscle soreness is not a requirement to make progress. Your goal is to effectively stress your muscles and nervous system across the training session, while still being able to recover before your next training session. This is how you continue to progressively overload your muscles each time you resistance train… and that’s how you make progress.

Remember, done right, resistance training is a very potent stressor. Usually stress has a negative connotation, but in the context of exercise, just the right amount of local and systemic stress is what will bring you results quickly and consistently. That’s why I designed this template with plenty of time for your muscles and connective tissues, and also your nervous system, to recover. Your goal is to trigger the proper adaptations by training really hard, then get out of the gym and recover while you live your life.

Don’t think “workout,” think “training”

First, training is, as the definition of the word implies, practicing. As Pavel Tsatsouline, world-renowned strength coach, explained on my podcast, it’s best to view and engage

in resistance training as if it were a self-taught class wherein you practice the skill of triggering specific adaptations.[79] The huge advantage of this approach is that it keeps you from adopting the "working out" mindset, which has most people merely going through the motions, thinking about burning more calories or the pump, or just getting through the workout. That mindset confuses effort with directed effort. And this template is all about directed effort designed to trigger specific adaptations. Calling it training carries a different level of seriousness that, in my experience, translates to higher quality effort and results. If someone asks "Training for what?" you can tell them, "to get stronger" or simply "life!"

Track your progress

Treating your resistance training as a self-taught class means you should track your progress. This helps you remember what you did last session so that you can make sure you're properly using progressive overload. I believe that whether you are a novice or highly experienced, you should take your training seriously—it's your body and your health after all.

Some people will track their sets and reps and the amount of weight they use on each exercise on their phone, others in a

[79] Andrew Huberman, "Pavel Tsatsouline: The Correct Way to Build Strength, Endurance & Flexibility at Any Age," Huberman Lab (podcast audio), February 10, 2025, https://www.hubermanlab.com/episode/pavel-tsatsouline-build-strength-endurance-flexibility-at-any-age.

training notebook, others using an app. It doesn't really matter, although the phone offers so many chances for distraction whereas a good ol' pen and some paper have far fewer.

I usually put my phone on airplane mode when I start my workouts and only use it for music, or I simply train without it. Some people like to listen to a podcast or book while they train. You decide what is best for you but, again, take your training seriously. Avoid texting and scrolling social media while you're training. Your workouts will likely be much more effective, and I can guarantee you they will be more efficient. I use a paper notebook to write down every set and number of reps I do in order to track my progress, as described above. This makes a tremendous difference because it gives you a specific goal to beat every exercise and set. It also makes it fun. You can strive for and see objective progress in almost every workout.

Many people are haphazard about their training. Thus, they get some results when they start, but their progress quickly evaporates. It takes the same amount of time (or even potentially less) to do a truly effective workout as it does to do a half-baked one. So make sure you do the former. The great news is that as you practice resistance training, you get better at it.

## BRIEF WHOLE-BODY WARM-UP: ABOUT 5 MINUTES

You'll get better results and avoid injury by including a brief five-minute whole-body warm-up at the start of each resistance training session. The goal is simply to increase your core body temperature a few degrees, to loosen up (a nontechnical term) your joints and to prepare for the sets to follow. I like to pedal on a stationary bike or walk briskly on a treadmill for five minutes until I break a light sweat. Alternatively, you can do jumping jacks, windmills, or get on the elliptical or stair climber. It doesn't really matter as long as you warm up. Physical therapist Kelly Starrett, PhD—famous for his work on mobility and all-around fitness—offered a great warm-up when he was a guest on my podcast, especially for when you're feeling less-than-motivated to train:

Just play for a few minutes. Shoot some layups and free throws on a basketball court, or shadow box, skip rope, or complete a simple circuit of body-weight-only exercises: first twenty-five air squats, followed by five lunges per side with just your body weight, and then jog in place for a minute before repeating, until five minutes have elapsed.[80]

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[80] Andrew Huberman, "Dr. Kelly Starrett: How to Improve Your Mobility, Posture & Flexibility," Huberman Lab (podcast audio), December 2024, https://www.hubermanlab.com/episode/dr-kelly-starrett-how-to-improve-your-mobility-posture-flexibility.

You may recall that bodily movement activates two circuits: one in the body and the other in the brain. These trigger the release of adrenaline and norepinephrine, respectively, which in turn makes you more physically energized and mentally alert.[81] That's the other reason to do a brief whole-body warm-up. Many people find they are dragging themselves to the gym, and, after warming up, they feel like a new person. Some of that is psychological, but there is a physiological component too—the way that whole body movement stimulates adrenaline (via the body-to-brain connection) "wakes up" your brain, as we discussed at the start of this chapter.[82]

## TOTAL NUMBER OF WARM-UP SETS: 1–3 SETS

Warm-up sets ensure that you are performing an exercise correctly across a full range of motion, and they prepare your nervous system, muscles, and connective tissues for the work sets that follow.

Most people do relatively high reps for their warm-up sets: For instance, they will do fifteen to twenty reps on the first warm-up set, add some resistance and do ten to fifteen reps

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[81] Zouhal et al., "Catecholamines and the Effects."

[82] Richard P. Dum et al., "Motor, Cognitive, and Affective Areas of the Cerebral Cortex Influence the Adrenal Medulla," Proceedings of the National Academy of Sciences 113, no. 35 (August 2016): 9922–27, https://doi.org/10.1073/pnas.1605044113.

more, then add a bit more resistance and do six to ten reps for the last warm-up set. Most people also tend to rest briefly between their warm-up sets, usually no more than thirty seconds.

I suggest doing warm-up sets differently, so you can get more out of your work sets that follow. Regardless of the repetition range you have planned for your subsequent work sets, make your first warm-up set only six to ten reps. Use a moderately light weight for you, around 40 percent of the amount of resistance you will use for your first work set. Lift and lower the weight under control throughout every set to make sure that you mentally engage the muscles you are trying to train. Then rest one to two minutes, and do a second warm-up set, but this time perform only four to eight reps using about 65 percent of the weight you'll use for your first work set. For some people that's enough warm-up before the first work set. If your work sets are going to be heavy on a given exercise, meaning they will have you reaching failure (that is, again, the inability to continue moving the weight in good form) after completing just five to seven reps, then I suggest you rest an additional one to two minutes after your second warm-up set and do a third and final warm-up set. For that third warm-up, use about 75 percent of the weight you'll use on your first work set. And for that third warm-up set, keep the reps low, to only two to four. Yes, you could do more, but that's not the goal. You're not trying to stimulate

any adaptations with your warm-up sets. Rather, think of them—and especially that third warm-up set—as rehearsals for the real deal: the work sets to follow. Warm-up sets are about preparing your body and mind for the work sets.

Note the percentages above are approximate percentages of the weight you plan to use for your first work set, not a percentage of your one-repetition maximum (1RM), or the maximum you can lift once in good form. Most importantly, your warm-up set should never come close to failure.

## TOTAL NUMBER OF WORK SETS PER MUSCLE GROUP: 4–9 SETS

To be clear: When I say muscle group I'm referring, for example, to your shoulders, or biceps, or hamstrings, or triceps, or quadriceps. "Arms" are not a muscle group. Nor are "legs." At least not in the context of resistance-training jargon. Quadriceps are called a "muscle group" because they are made up of several muscles. The same applies to your biceps and all the rest. Later in the chapter I provide links to exercise videos that will clarify any confusion about this and what exercises to select when training. Doing four to nine work sets per muscle group (divided across two to three different exercises) is more than enough to effectively trigger the specific adaptations you want, if you focus and work hard to control the resistance throughout every rep of

every work set until you can't move the weight any more, aka failure.

So should you do four or eight, or five, or seven sets?

You decide, but be logical about it based on your goals and needs. One thing that is rarely discussed but important in designing the best program for you is that many people have muscle groups that, for a variety of reasons related to their genetics, the sports they played as youth, or their resistance training history, respond much faster to resistance training than other muscle groups. It makes sense to do fewer total work sets for those muscle groups. You heard me. Do fewer work sets for your quicker-to-respond muscle groups. That will ensure you don't throw off your proportions and create strength imbalances that eventually will lead to injuries. Women seem to be much better about training for aesthetic balance than men. We have all seen "that guy"—the one with the big chest and arms but disproportionately much smaller legs, or the one whose shoulders and back are as wide as barn doors, but with a tiny neck that makes it look like his head is on the wrong body. It's not just about aesthetics; muscle strength and size imbalances set you up for injuries in and out of the gym.

Aim for strength and aesthetic balance. If you have a muscle group that responds easily, limit yourself to doing four to five or even just three work sets total for that muscle group

in a given workout. You should still focus on training to failure and in great form during those sets. For all other muscle groups, I suggest you do four to eight total work sets.

One last point regarding the total number of work sets to do. The better you focus and contract a given muscle group, the fewer work sets it will take to stimulate the specific adaptations you're after. Think about it this way: If you've ever hammered a nail into wood, you know that one to two clean, well-placed strikes of the hammer drives the nail in flush. If you don't hit the nail squarely on the head, however, you need many more swings. When you do four to eight work sets per muscle group, and just two to three work sets per exercise, focusing from start to finish as you lift and lower the resistance, your task of triggering specific adaptations in that muscle group is complete. Additional sets are not needed and might do more damage than good—just as continuing to pound on a nail that is already hammered in simply damages the wood.

How do you measure your neuromuscular control? Jeff Cavaliere, a physical therapist and strength and conditioning coach who worked with the New York Mets before founding his own online exercise platform, AthleanX, has a cool trick that translates to more effective (adaptation-stimulating) workouts too. Can you contract a muscle, let's say your bicep or calf or quadricep hard enough to make it almost feel

like a cramp but with no added resistance, just by “flexing” the target muscle? If so, that is good news because when you train using isolation exercises, that’s effectively what you are trying to do. You’re just adding resistance to contract the muscle even harder, throughout the entire movement. This one takes some time for people to wrap their heads around because everyone thinks about “lifting weights,” but effective resistance training aimed at generating hypertrophy and strength improvements is about challenging muscles, and the weight or resistance is just a tool to allow you to do that more effectively.

Compound exercises challenge specific groups of muscles and also train the coordinated action of those muscles as they work together to perform certain movements. A squat, for instance, calls on the muscles of the quadriceps, hamstrings (somewhat), glutes and lower back, and numerous stabilizer muscles. “Feeling” your quadriceps during a squat is crucial but the expectation is not that you completely isolate them; in fact, that’s impossible. That is why in this program I suggest doing both compound and isolation exercises.

And if you can’t feel the muscle you’re trying to target when doing compound exercises, consider doing the isolation exercise for that muscle group first, then the compound exercise for that muscle group. For example, if you are having a hard time feeling your chest (pectoralis) muscles

during your bench presses, do some sort of chest-isolation exercise first, then move to your bench presses. You don't have to move immediately from one to the other in classic "pre-exhaustion" format. You can simply do your one to three work sets of a chest-isolation movement and then do your one to three work sets of bench presses or other compound movement for the chest.

By the way, if you're feeling lost with these mentions of exercise names or anything else, do not worry! I provide numerous examples of isolation and compound exercises at protocolsbook.com/videos. You'll of course have to use less weight on the compound movement if you do an isolation movement for the same muscle group first, but remember why you are there in the first place: to trigger specific adaptations in specific muscle groups. And you can always reverse the order and proceed with the compound movement a few months later, once you have established a strong mind-muscle-connection with that formerly hard-to-feel and hard-to-improve muscle group.

Is it possible to do too few sets? Yes.[83] Or I should say, most trainers would say yes. Arthur Jones and Mike Mentzer were famous for taking the low-volume–high-intensity logic to the

[83] James W. Krieger, "Single vs. Multiple Sets of Resistance Exercise for Muscle Hypertrophy: A Meta-Analysis," Journal of Strength and Conditioning Research 24, no. 4 (April 2010): 1150–59, https://doi.org/10.1519/JSC.0b013e3181d4d436.

extreme, arguing that people should do just one to two work sets per muscle group per week. That was their suggestion: not per exercise but across the entire workout! Most everyone agrees nowadays (and the science does too) that you need more work sets. Studies over the last two decades reinforce the idea that somewhere more in the range of four to eight work sets per muscle group is both necessary and sufficient to trigger near maximal strength and/or hypertrophy adaptations.[84]

Many people try to train each muscle group twice per week, with twelve to twenty work sets per group at each session, believing that this is the "optimal" frequency for stimulating muscle growth. But believe me: Doing fewer sets with more intensity, with the focus described here, three times per week, allows for a more sustainable practice and thus better results over time. I will die on this hill! And again, the goal

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[84] Mark D. Peterson et al., "Maximizing Strength Development in Athletes: A Meta-Analysis to Determine the Dose-Response Relationship," Journal of Strength and Conditioning Research 18, no. 2 (2004): 377–82, https://doi.org/10.1519/00124278-200405000-00031; Brad J. Schoenfeld et al., "Dose-Response Relationship Between Weekly Resistance Training Volume and Increases in Muscle Mass," Journal of Sports Sciences 35, no. 11 (June 2017): 1073–82, https://doi.org/10.1080/02640414.2016.1210197; Baz-Valle et al., "Systematic Review of the Effects"; Roberto Bernzárdez-Vázquez et al., "Resistance Training Variables for Optimization of Muscle Hypertrophy: An Umbrella Review," Frontiers in Sports and Active Living 4 (July 2022): 949021, https://doi.org/10.3389/fspor.2022.949021; Brad Jon Schoenfeld et al., "How Many Times per Week Should a Muscle Be Trained to Maximize Muscle Hypertrophy? A Systematic Review and Meta-Analysis," Journal of Sports Sciences 37, no. 11 (2019): 1286–95, https://doi.org/10.1080/02640414.2018.1555906.

is to do cardiovascular workouts three times a week, too, some of which are intense. That mix is essential for health and is covered in Exercise Protocol 3.

And the great part about this program is that you still end up targeting each muscle group twice per week, just not the way you might think. Let me explain.

As I mentioned above, with regard to frequency of training, you can find studies that argue training each muscle group two or even three times per week is best for these results.[85] I suggest resistance training for each major muscle group directly only once per week. This is not by accident. By scheduling this way along with specific types of cardiovascular training sessions on particular days, you actually end up training each muscle group twice per week: once directly, and once indirectly. On chest and back day, your shoulders and arms get some indirect training. On shoulders and arms day, your chest and back get some indirect training too—because you are doing compound movements like overhead presses and dips and perhaps also some palms-facing-you pull-ups for your biceps. On "sprint" cardio day (covered in Protocol 3), your legs get indirect stimulation, and if you do your sprints on an assault bike, those stationary bikes with a fan and handles that you pump

[85] Helene Pedersen et al., "High-Frequency Resistance Training Improves Maximal Lower-Limb Strength More Than Low Frequency," European Journal of Sport Science 24, no. 5 (2024): 557–65, https://doi.org/10.1002/ejsc.12055.

forward and back while you pedal, your back and arms get some additional indirect training too. Of course, these indirect training sessions are not enough to stimulate maximal strength or hypertrophy increases on their own, but they synergize beautifully with the direct resistance training that you're already doing for a total of three days a week with weights. A sprint four days after a proper leg workout for calves, quadriceps, hamstrings, and glutes, with multiple sets taken to failure… you'll be able to do that, I promise, but it's right as your legs start to feel fresh again… and a few days later you're back to your leg workout. You won't lose ground. In fact, you'll come out ahead on both. Try it.

In my experience, the overall program in this chapter provides a well-rounded combination of strength, hypertrophy, endurance, and flexibility and is the best time investment to achieve the results most people are seeking. It also keeps your motivation level for each workout high, which is crucial for anyone leading a busy life. Going to the gym for hard workouts five days a week is difficult for most to maintain. Pretty soon, people are downing two energy drinks and a coffee just to get going, not sleeping well, and getting sick. But three days a week, with just two to three hard work sets per exercise and two to three exercises per muscle group and three or four muscle groups? That's something I'm confident you can do for the rest of your life—and that's how to think of both getting and staying in

great shape. It's for all these reasons that I strongly believe in this program. Let me put it this way: I've given this program to men and women of a wide range of ages, athletic backgrounds, and health histories, and I've never seen it not work for them to reach their fitness (and aesthetic) goals.

## TOTAL NUMBER OF EXERCISES PER MUSCLE GROUP: 2–3 EXERCISES TOTAL NUMBER OF WORK SETS PER EXERCISE: 2–3 SETS (USUALLY JUST 2)

In most cases, doing two exercises that target each muscle group is plenty, but in some cases you may want to do a third exercise if it's a muscle group of particular focus. Once again and so that we're clear on terms: Legs are not one muscle group. Quadriceps are. Hamstrings are another muscle group. Your legs have quadriceps, hamstrings, glutes, and calves—each of which has its own subdivisions. You do two to three exercises for each of these groups.

It's crucial that at least one of the two to three exercises for a muscle group be a compound (multi-joint) exercise. In most cases you'll also want to do one isolation (single joint) exercise for the same muscle group. Doing two to three exercises (with two to three work sets for each exercise) brings you to four to nine work sets per muscle group. (I recommend not going above nine work sets total for a single muscle group in a workout, given that you are training

multiple muscle groups in each workout. Indeed, aiming for six hard work sets per muscle group per workout is often better, provided that at least three of those sets are from a compound exercise and you take those work sets to failure.)

Time constraints will often dictate exactly how much you do. So will how well you slept (I suggest aiming for the lower end of the number of sets on days when you are a bit sleep-deprived, but still doing the compound movements). So will nutrition. And stress. If you're well rested, fed, and you have the time, do the full three exercises for two to three work sets each, for six to nine total work sets. Or just do four to six work sets per muscle group total and pour every bit of focus and energy into those. With time, you'll learn when enough is enough for that muscle group on a given day and according to the context of the rest of your life.

## Specific exercise selection

As mentioned earlier, target a muscle group with at least one compound exercise and one isolation exercise for that same muscle group. (I provide a set of video tutorials and options to select from at protocolsbook.com/videos).

The rationale for including one of each is that compound exercises target a given muscle in conjunction with other muscles that it normally works with in various everyday movements. That also means compound movements allow

you to use heavier weights, which provide not just a direct effect on the muscles you're training but also a positive "systemic effect" on various hormones and neurotransmitter systems of the brain and body.[86] And because they mimic a lot of pushing, pulling, kneeling, and lunging movements we do in everyday life, compound movements carry over to "real world" strength. The ones done standing, like squats and dead lifts, overhead presses, and even bicep curls or overhead extensions, help in developing more core stability and strength too. Taken to failure for two to three well-controlled, focused sets per muscle group, compound exercises are great… but they are pretty taxing on your nervous system. This is especially important considering you are doing at least one such movement, and perhaps even two, per muscle group, and training multiple muscle groups per workout.

Follow with an isolation exercise for each major muscle group. It lets you intensely target a specific muscle, it

[86] Kraemer and Ratamess, "Hormonal Responses and Adaptations"; William J. Kraemer et al., "Effects of Heavy-Resistance Training on Hormonal Response Patterns in Younger vs. Older Men," Journal of Applied Physiology 87, no. 3 (1999), https://journals.physiology.org/doi/full/10.1152/jappl.1999.87.3.982; William J. Kraemer et al., "Recovery Responses of Testosterone, Growth Hormone, and IGF-1 After Resistance Exercise," Journal of Applied Physiology 122, no. 3 (2017): 549–58, https://journals.physiology.org/doi/full/10.1152/japplphysiol.00599.2016; Simon Haunhorst et al., "Acute Effects of Heavy Resistance Exercise on Biomarkers of Neuroendocrine-Immune Regulation in Healthy Adults: A Systematic Review," Exercise Immunology Review 28 (2022): 36–52, PMID: 35452397.

reinforces the mind-muscle connection to that muscle, and, because of the mechanics of the movement (the position and range of motion, etc.), isolation exercises tend to develop aspects of the target muscle that compound movements may not target as thoroughly.

And they don't create quite as much overall stress on your nervous system. Yes, exercise-induced stress is good (!) but it adds up and, as stated earlier, you have just one nervous system for both your workouts and for the other aspects of your life. You don't want to overtax it. That does not mean go easy on isolation exercises. You still need to use progressive overload and to take those isolation exercise sets to failure. By the way, once you do two work sets of leg extensions or side lateral raises (an isolation movement for the shoulders) for twelve to fifteen well-controlled reps to true failure, you'll realize that isolation movements done properly are their own special kind of intense!

To illustrate the importance of exercise and complementarity, let's return to that earlier example and talk about training your quadriceps. You're choosing two exercises. The compound exercise could be a barbell back (or front) squat (your hip and knee joints both move in a squat; hence, multi-joint), and the isolation exercise could be the leg extension machine (only your knee joint moves and your hips stay locked). If you have the energy that day, you

might also do another two to three work sets of a compound exercise for your quadriceps, such as leg presses or hack squats.

As you likely realize by now, the exercises you can select from are nearly endless. And they can be free weight or you can use weight machines, or even just your body weight for some of them, like pull-ups or barbell dips. What's important is that you can do a movement safely, without pain, and while still pushing yourself to failure. And that you can safely add more and more resistance to the movement over time as you get stronger. People debate exercise selection a lot. Assuming you include both compound and isolation movements, you have to find what is best for you. What equipment you have access to is of course also a factor. Again, the video links I provide give you lots of options. In the meantime, let's discuss the "best exercise" myth.

## The "best exercise/movement" myth

Barbell squats (where the bar is on your shoulders as you squat down and stand up again) are a terrific compound exercise many people can and should use to target their quadriceps, glutes, and other leg muscles. They challenge your lower back and core muscles too. Some trainers will say that you absolutely must do them, but that's simply not true. Some people have hip, leg, or shoulder mechanics that prevent them from doing barbell back squats without joint

pain during the exercise or in the days after. You could work with a trainer to overcome those issues, but you could also substitute front squats, where the bar is held on the front of your shoulders, or hack machine squats, which have your back on an angled sled for support, or even leg presses for your quadriceps instead. There are Zercher squats, rear-leg-elevated squats, and goblet squats. There are also a near-endless number of lunge variations that target the quadriceps, glutes, and other muscles emphasized in the barbell back squat. The internet is filled with exercise clips showing all of these exercises. Again, I provide video tutorials of many to choose from at protocolsbook.com/videos.

Find the exercises you can really push yourself in, without joint pain. You'll enjoy it more, and your results will be greater… which will lead you to enjoy it even more. You'll experience a whole different relationship to resistance training if you make yourself the CEO of the process. Only do exercises that really work for your body. Just don't only do isolation movements. Compound movements are necessary to get the full benefits of resistance training.

## A few quick notes about safety and pain

Safety: I'll be direct about this. Be careful as you train. Done correctly (warm up, don't add more weight than you can handle with control, and so on) and with caution to avoid a

few potentially perilous mistakes, resistance training as described in this protocol is a very safe activity.

What sort of perilous mistakes am I referring to? Well, people have died barbell bench-pressing.[87] They couldn't push the bar up from their chest and they asphyxiated. It's rare, but it happens. Have a spotter who knows how to spot you correctly—which means they have to be strong enough to pull the loaded bar up off your chest or neck if need be. If you don't have someone who can do that, an easy solution is to use dumbbells if you're bench-pressing alone—those can be set down to your knees and dropped to your sides if you hit failure. A few dents in the floor are far better than the alternative. Or use a rack with safety pins (sturdy bars that can support a loaded bar) to set the weight on if you hit failure mid-repetition. Same goes for barbell squats. Yes, you can "dump" the bar behind you if you hit failure, but far better to have a knowledgeable spotter. If no such person is around, again use a rack with pins so you can set the bar down and crawl out. Many people opt for machines like the leg press or hack squat for compound quadriceps movements for just this reason: It's easier to hit failure without doing what I described. And they usually have "stops" you can engage when you hit failure. No, these movements do not

[87] Mary I. Jumbelic, "Traumatic Asphyxia in Weightlifters," Journal of Forensic Sciences 52, no. 3 (April 2007): 702–5, https://doi.org/10.1111/j.1556-4029.2007.00429.x.

engage the same muscles to the same degree as free weight compound exercises. Yes, they are safer. You decide your risk tolerance, but be safe regardless.

Also, think about your posture every time you are picking up or putting down weights, even if you're just replacing them on the rack or getting in position to start your set. A lot of resistance training injuries are the consequence of people not being mindful as they do the in-between work sets stuff because their mind is someplace else.

I'm proud of the fact that I have been training hard, and heavy (for me) for nearly thirty-five years and have never had a serious injury from resistance training. (Knock on wood!) Sure, I've had a few minor strains here and there but nothing that floored me or kept me from training or daily activities for more than a few days. I attribute that to staying mentally focused from start to finish in the gym, and to only making relatively small incremental increases (5 to 10 percent) in the amount of weight I add to any given movement from week to week, and only once I can do at least eight to twelve reps of that movement in good form.

Pain: We have all heard the mantra "no pain, no gain." News flash (!): That does not refer to joint or connective tissue pain. It refers to the burning sensation in muscles caused primarily by the accumulation of hydrogen ions inside muscle cells, brought on by the fact that the muscle cells

need energy faster than oxygen can be delivered to them. (No, not lactic acid as was once believed. Lactate is created, but we know now it is not the cause of the burning.)[88] If an exercise causes sharp or dull joint pain—not post exercise soreness but, again, pain—either your form or the exercise itself is not right. Don't persist with that exercise. At least not that day. Find an alternative.

## REPETITION ("REPS") RANGE PER SET

When it comes to selecting the number of reps to do for your work sets, you have options. After decades of debate about "optimal rep ranges," research from Brad Schoenfeld's lab, at Lehman College, found that using repetition ranges of anywhere from five all the way up to thirty on work sets stimulates essentially the same amount of hypertrophy. That is, if (and only if!) those sets are taken to failure.[89] Generally speaking, this is great news because it means that if you select a weight that you think you'll do eight reps with (and fail on rep nine) but somehow you are able to do eleven reps (and fail on rep twelve), you'll still make progress. It removes the need to stay in a narrow rep range, at least for hypertrophy.

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[88] Simeon P. Cairns, "Lactic Acid and Exercise Performance," Sports Medicine 36, no. 4 (2006): 279–91.

[89] Brad J. Schoenfeld et al., "Strength and Hypertrophy Adaptations between Low- vs. High-Load Resistance Training," Journal of Strength and Conditioning Research 31, no. 12 (December 2017): 3508–23, https://doi.org/10.1519/jsc.0000000000002200.

At face value, these findings suggest you don't ever need to train with low reps and heavy weights to get results; you simply need to push until the final reps of your work set are very hard to complete. In some sense, yes, you can use moderate or even light weights (for you) that limit you to twelve to thirty reps before reaching failure. But you should also regularly use heavy (for you) weights, which means weights that limit you to just five to eight reps and no more, despite your maximum effort to complete more. Turns out using a mix of weights stimulates different muscle fiber types and ensures you activate different forms of cellular stress and thus adaptations.[90]

More on that below but in the meantime, if you don't want to have to think much about rep range selection, here's a good guide:

For compound exercises, use the amount of weight on your work sets that has you reaching failure between five and twelve reps.

- Occasionally use heavier loads that limit you to just 3–5 reps before you reach failure.
- Experiment with lighter loads that allow you to complete higher repetition compound exercises, especially for

[90] Brad J. Schoenfeld et al., "Differential Effects of Heavy Versus Moderate Loads on Measures of Strength and Hypertrophy In Resistance-Trained Men," Journal of Sports Science and Medicine 15, no. 4 (December 2016): 715–22.

quadriceps and calves. Many people find that 12–20 repetition squats (or its variations) are very effective for hypertrophy.

- Warning: High-repetition quadriceps work sets taken to failure are brutal. The loads you use will be lighter than you would use for 5–10 reps taken to failure, but the pain tolerance level on the final reps (and mental fortitude) required for 15–25 reps of a compound leg movement, like squats or even one of its machine variations, like a hack squat, taken to failure is no joke.

For isolation exercises, use an amount of weight on your work sets that has you reaching failure between eight and fifteen reps.

- Doing isolation exercises for heavier 5-8 repetition sets (to failure) is not bad per se, but most people find that doing higher (8–15, or as many as 25) reps on isolation movements like leg extensions, shoulder lateral raises, preacher curls for biceps, and so on, is more effective. But feel free to experiment. You may find, for instance, that your hamstrings respond best to heavier sets of lying leg curls (an isolation exercise) that limit you to 5-7 reps.

Just remember, no matter what the repetition range you use for a movement, go to failure on work sets, even if it has you exceeding your target repetition range for that day and exercise.

Now before you say, “I heard leaving one to three ‘reps in reserve’ (RIR) is just as good as training to failure,” we need to get honest: If you use heavy (for you) weights that would make you reach failure somewhere between five and ten reps, but you stop a rep or two before failure, you probably will still stimulate some strength and hypertrophy adaptations. That is because heavier loads generate levels of mechanical tension that readily stimulate muscle and nerve-to-muscle adaptations.[91] However, given that this program has you doing relatively fewer work sets (four to nine), and directly targeting each major muscle group only once per week (and a second time per week, indirectly, as described above), you really should take all your work sets to failure.

Still not convinced? Well, consider this reality: We all like to think we are training our absolute hardest, but what most people consider taking a set to failure almost always leaves one to two more repetitions in reserve. How can I be so sure? Well, if you hit failure on the eighth rep of a set but were told that getting a ninth rep would save someone’s life, I am willing to bet that ninth rep would happen. So failure is somewhat subjective. Thankfully lives are not on the line

[91] Brad S. Currier et al., “American College of Sports Medicine Position Stand. Resistance Training Prescription for Muscle Function, Hypertrophy, and Physical Performance in Healthy Adults: An Overview of Reviews,” Medicine and Science in Sports and Exercise 58, no. 4 (March 2026): 851, https://doi.org/10.1249/MSS.0000000000003897.

according to when we reach failure on our lifts. But that momentary muscular failure is the closest thing you have to an objective measure of intensity of effort. So seek it out.

Here's a bit more information so you can make the best-informed decisions about whether to train heavy, moderate, or light (for you). Hypertrophy adaptations are thought to be stimulated by at least three things:[92]

- 1 Microdamage caused to the muscles using heavier loads. (Here, we can define "heavier loads" as sets that have you fail after completing three to six reps.) The extent to which this contributes to hypertrophy is a matter of emerging debate, but it still has some solid evidence in its favor.
- 2 Mechanical tension (from moderate to heavy loads and/or the contraction or stretching of the target muscle).
- 3 Metabolic stress (including the pump and cellular swelling, all brought on by enhanced blood flow, along with the release and delivery of various cellular factors to the muscles you are training).

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[92] Brad J. Schoenfeld, "The Mechanisms of Muscle Hypertrophy and Their Application to Resistance Training," Journal of Strength and Conditioning Research 24, no. 10 (October 2010): 2857–72, https://doi.org/10.1519/JSC.0b013e3181e840f3; Henning Wackerhage et al., "Stimuli and Sensors That Initiate Skeletal Muscle Hypertrophy Following Resistance Exercise," Journal of Applied Physiology 126, no. 1 (2019): 30–43, https://doi.org/10.1152/japplphysiol.00685.20; Carl Juneau, "Training for Hypertrophy: The Case Against Muscle Damage," Stronger by Science, accessed May 20, 2026, https://www.strongerbyscience.com/muscle-damage.

Muscle size and strength are correlated so if you make a muscle bigger it will get stronger, but training specifically to maximize strength is typically done by using loads that limit you to three to five reps (or fewer) and doing a large number of sets for each muscle across the week and not training to failure on any one set. Getting strong is great but the vast majority of people who resistance train do so to get stronger and for aesthetic changes that are best brought about by the rep and load ranges discussed above.

That is why I suggest you mainly use five to twelve reps on your compound work sets, and occasionally venture into using heavier loads that limit you to just three to five reps before you fail. In reality there will be times when you can do ten reps of a movement with a given weight, so you increase the amount of weight the next time you're training that body part and somehow you can only do three or four reps before you reach failure. Don't worry about it. Just keep striving to add more reps or weight each workout in great form. If you want to obsess over something useful, obsess over the amount of focused effort you put into your work sets and using great form!

About ten years ago, I decided to train exclusively with heavy weights that limited me to three to five reps on compound exercises and six to ten reps on isolation exercises—which is called periodization. I did that for three

months and I got a lot stronger, such that when I went back to doing five to twelve reps, I was stronger on every movement. Later I ran a similar experiment using lighter weights and more reps (12–25) for a few months… all still to failure on work sets of course. I made considerable progress in terms of hypertrophy, but when I went back to using heavier weights for lower reps, I found I had lost a lot of ground. So I learned something. Nowadays I almost always train with weights that limit me to five to twelve reps and only venture up to fifteen to twenty for an especially intense leg workout or on certain upper body isolation exercises. That's what is best for me and my goals. You should experiment and find what is best for you.

## Progressive overload is key!

I know that the requirement for progressive overload is a giant "duh" to many of you out there, but others may need this reminder. If you do the same workout every week, using the same weights, reps, and sets, take note. You might sweat, work hard, and feel the mood enhancements of exercise but your progress will halt. Because we're keeping the work sets to just two or three, you must challenge your muscles by either increasing the amount of weight or the number of reps, or both, while keeping your form excellent.

"More weight or more reps, perfect form, and pushing to failure." That should be your resistance training mantra!

Be aware that it's unreasonable to expect to add weight while maintaining the same rep count you performed the previous week, but it will happen. Those are great workouts. However, there will be weeks when, for whatever reason, you are not able to hit your target weight and rep goals in the gym. Consider the reasons: Are you sleep-deprived? Underfed, under caffeinated? Unfocused? Life happens, so don't obsess about it. Just do your absolute best for that day. If you find you're not making progress for two weeks in a row, you may need to reduce the set volume. Yes, I said it: Try doing fewer work sets and see if that budges you forward. Or maybe it's time for a week off to rest, so you can come back stronger.

Finally, while 1RM lifts and even 2RM lifts are great for building strength (and allow you to answer the "How much can you bench?" question you'll invariably get at some point), I recommend most people avoid doing 1RMs. They come with too much risk of injury. If you really want to do them, fine, but use a spotter, please! And understand that 1RM lifts are what you train for in powerlifting—which is a specific sport. If that's what you're interested in, there are great programs out there for powerlifting. But most people are lifting weights for other reasons. They want the variety of benefits that come from the resistance training I've described here. If someone ever asks me how much I can

bench, I usually just tell them what I did for five to eight reps the preceding week because that's all I know for certain!

## An important note about training to, and past, failure

As I've said about a hundred times now (but still a hundred fewer than Mentzer would have in the same time frame!): Do the first work set of an exercise to failure, using perfect form. Pour everything you have into that set, on every repetition. That means, as the final reps approach, think: "This is what stimulates the adaptations I'm here for." Strive to keep your form stricter, and stress the targeted muscles more, not less as failure approaches. Then, when the weight won't move any more, try HARD to continue to move it for an additional three to five seconds, then re-rack the weight.

Then rest two to three minutes, or perhaps even four to five minutes on compound exercises for legs or back or other large muscles. Then do a second set of that exercise, also to failure, but don't fight to push the weight once you've hit failure, like I described for the first work set above. This is not a back-off set. You're just stopping when you can't move the weight anymore.

Should you do a third set of that exercise? Maybe. But usually not. Two sets done the way I just described should trigger the specific adaptations you're seeking. And you still have another exercise or two to complete next for that

muscle group. Am I like Mentzer and think that if you do a third set, it's going to be counterproductive? No. I don't know if it will help your progress, but I don't think it will hurt you if done in good form and with intense effort. You decide. One thing's for sure: If you can't make the set a highly focused, max effort work set, move on.

Once you hit failure, should you do assisted (sometimes called "forced") reps? Or a drop set, where you reach failure then quickly lower the amount of weight and grind out one to three more reps before hitting failure again? These techniques can be helpful, but they stress your nervous system a lot. I would suggest pushing beyond failure for two to three sets maximum across the entire workout. On days where you're feeling a bit stressed, don't push past failure. And if you have someone assisting you with forced reps, make sure they don't jump in before you reach failure. Make sure they force you to do the work. It's a fine line between spotting someone to ensure they don't get stuck in a dangerous position during a lift and getting someone to help you with forced reps.

Spotting someone in the weight room is a skill—it's not brain surgery but there is a right and a wrong way to do it. On social media, you often come across videos of gym disasters, where people get stuck under the bench press or squat bar because they were too shy to ask for a spot. There

are, however, a nearly equal number of gym disaster videos of young kids who were not able to pull the bar off their dad's chest because they are too weak. Avoid all of this by making sure anyone you ask to help you is (1) strong enough to help with the amount of weight you're using and (2) understands that you need to do the majority of the work. They are there to help keep you safe (again, especially on bench presses or squats where getting stuck can be dangerous), but not hinder your progress. Tell them: "I'll let you know when to help but don't touch the weight until I say." And if they check their phone during your work sets, find a new training partner!

## REST DURATION BETWEEN SETS

Just as I listed above, take one to two minutes off between warm-up sets, and two to five minutes between work sets, depending on repetition range. Typically, people rest longer (3–5 minutes) between work sets of compound exercises than between work sets of isolation exercises (90 seconds to 3 minutes). Some people adhere to the clock for between set rest, others go by feel. You decide. I use the clock to get a sense of how my subjective "Okay, I'm ready for another set" sense matches up with various time ranges. Just don't talk, text, or scroll too much between sets. Stay focused, get the training done and get out.

## Recovery tips

Then there's the rest that comes after your resistance training session.

Great things happen for your health during your resistance training sessions but the really big benefits—the actual changes in your muscles, connective tissues, and brain—occur during the two to four days afterward, mostly while you sleep.[93] The protocols provided elsewhere in this book will help you get your sleep right and your nutrition and stress control right—all of which support recovery from, and adaptation to, exercise. You do not need to overhaul your sleep schedule or diet just because you resistance train. It is true that if you do a particularly hard workout, you may find you need a bit more sleep on subsequent nights, but in general, resistance training has the effect of improving the quality of your sleep, not increasing your sleep duration needs.

Same with nutrition. Resistance training doesn't massively change your nutritional needs. Assuming you are already eating in the manner I describe in chapter 4—meaning you strive to eat about 1 gram of quality protein per pound of

[93] A. Chesley et al., "Changes in Human Muscle Protein Synthesis After Resistance Exercise," Journal of Applied Physiology 73, no. 4 (1992): 1383–88, https://doi.org/10.1152/jappl.1992.73.4.1383; J. Duncan MacDougall et al., "The Time Course for Elevated Muscle Protein Synthesis Following Heavy Resistance Exercise," Canadian Journal of Applied Physiology 20, no. 4 (1995): 480–86, https://doi.org/10.1139/h95-038.

lean body weight daily, as well as plenty of vegetables, some fruit, unprocessed starches, and healthy fats.

It is true that if you are actively seeking to gain muscle, you'll need to eat more calories than you burn each day, but not by much—200 to 400 calories more than your maintenance calories is fine.[94] The old-school practice of bulking up by eating many more calories than you need is just not healthy—and brings with it a lot of body fat that you'll be trying to lose later.[95] I suggest you don't go that route. Provided you are not undernourished in calories or protein, you'll make progress with resistance training. Keep in mind as you gain more muscle and/or lose fat, your caloric needs change—fortunately in a way that lets you eat the same or more while staying lean or even getting leaner. Muscle tissue is metabolically demanding, which means it burns more calories than fat tissue, which is just one of the many benefits of resistance training.

There is a formula for determining your maintenance calorie needs per day, or what is called your basal metabolic rate (BMR). That's how many calories you should eat each day

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[94] Eric R. Helms et al., "Effect of Small and Large Energy Surpluses on Strength, Muscle, and Skinfold Thickness in Resistance-Trained Individuals: A Parallel Groups Design," Sports Medicine—Open 9, no. 1 (November 2023), https://doi.org/10.1186/s40798-023-00651-y.

[95] Gary John Slater et al., "Is an Energy Surplus Required to Maximize Skeletal Muscle Hypertrophy Associated with Resistance Training," Frontiers in Nutrition 6 (2019), https://doi.org/10.3389/fnut.2019.00131.

to maintain your current body weight if you did no physical activity whatsoever.

The formula is called the Katch-McArdle formula.[96] It takes about five minutes to figure out, and it's very easy to do. It requires a few other simple calculations first, which I'll walk you through.

Determine your body weight (BW) in kilograms:

- The formula uses kilograms so if you're used to thinking about weight in pounds and not kilograms, first divide your body weight (BW) in pounds by 2.2 to get your weight in kilograms.
- For example, I weigh 220 pounds, so 220/2.2 = 100 kilograms.

Calculate your lean body mass (LBM):

- Multiply your BW in kilograms by 1 minus your body fat percentage as a decimal.
- For example, I weigh 220 pounds (100 kg) and am about 10% body fat, so my LBM is approximately 100 × (1 – 0.10) = 90 kilograms.

[96] F. I. Katch and W. D. McArdle, "Validity of Body Composition Prediction Equations for College Men and Women," American Journal of Clinical Nutrition 28, no. 2 (1975): 105–9, https://doi.org/10.1093/ajcn/28.2.105.

- To estimate your body fat percentage, these online resources can help you figure that out: Men: https://athleanx.com/articles/body-fat-percentage-men
- Women: https://ultimateperformance.com/your-goal/fat-loss/female-fat-loss/womens-body-fat-percentage-in-pictures/
- There are more precise methods to measure body fat percentage, but they can be costly. This works pretty well, and is available at no cost.

Calculate your basal metabolic rate (BMR):

- Katch-McArdle equation: BMR = 370 + (21.6 × LBM in kilograms).
- So for me: BMR = 370 + (21.6 × 90) = 370 + (1944) = 2,314 calories per day, my BMR.

Multiply your BMR by an activity multiplier to account for how active you are throughout the day:

- Sedentary (no exercise at all): BMR × 1.2
- Lightly active (e.g., walking 7,000–10,000 steps): BMR × 1.375
- Moderately active (e.g., same as lightly active, plus yoga, “easy” cardio): BMR × 1.55
- Very active (e.g., walking 7K–10K steps, effortful resistance training or cardio): BMR × 1.725

- Extremely active (e.g., same as very active but two-a-day-workouts): BMR × 1.9

I'll add one comment on estimating your activity levels from sports performance expert Andy Galpin. Based on his experience training hundreds of clients in reaching their fitness goals, Andy notes that people are typically far less active than they think. For example, most who rate themselves "very active" are actually moderately active. He recommends that people select the activity category below the one they think they fall into if they're interested in losing weight.

I don't do resistance training and cardio on the same days, but I try to get seven thousand to ten thousand steps in every day, and I resistance train three times a week and do some form of cardio three days a week.

Occasionally, I will resistance train or run, and then also go for a hike. That constitutes extremely active but it's too rare an occurrence. Also, given Andy's recommendation, I'll consider myself "very active" and therefore multiply my basal metabolic maintenance calories (2,314 calories) by 1.725 = 3,991.65 calories… and I round that up to 4,000 calories per day.

I naturally tend to eat less on days I don't exercise, so for that day, I multiply 2,314 by 1.375 = 3,181.75 calories.

Voila! I need about 4,000 calories per day on my training days and about 3,000 calories per day on my non-training days. Do I neurotically count every calorie? No. Truth be told, I probably eat about 3,500 calories on training days, and about 3,000 calories on days I don't train.

Should you count calories? Maybe, but not every day and certainly not forever. I suggest figuring out your basal metabolic maintenance calories, what you're consuming now, and what changes you need to make, and then stick to that revised plan most of the time.

After you've completed the simple formula above, a great tool to track what you're eating is the Carbon app developed by Layne Norton, PhD, who is an expert in training and nutrition. (Carbon is a sponsor of my podcast because I believe it to be a terrific product.) Another option is the Cal AI app. Both make it very easy to enter what you eat and the apps then do all the calculations of calories, as well as macronutrients like protein, fats, and carbohydrates. Each carries a nominal cost, but it is very easy to use. Personally, I don't like to count calories every day, but I use the Carbon app for periods of time to see how I'm doing.

If you want to lose weight, eating 300 to 500 fewer calories than you burn daily is the best way to go. By resistance training while in this caloric deficit, you're better able to

spare lean muscle and preferentially lose fat.[97] This is further helped by making sure you're eating a protein-rich diet, despite eating 300 to 500 fewer calories. In short, resistance training and a good diet are two of the most effective and durable paths to body recomposition—a fancy term for "gaining muscle and losing fat," as opposed to simply losing weight.

## Pre-training and post-training nutrition

I prefer to resistance train early in the morning or midafternoon, and I consistently have my best workouts when I have not eaten anything in the hours before I begin. I like to wake up, get sunlight, hydrate, caffeinate, then train, and then eat afterward, but some people just can't do that. They feel weak, tired, hungry, or nauseous if they train on an empty stomach. But I know myself well. If 10:00 or 11:00 a.m. rolls around and I haven't eaten, I need to ingest some calories before training. That's usually a few scoops of whey protein, a small bowl of oatmeal, a piece of fruit, and some nuts. Nothing fancy. Or two to three eggs, instead of the whey. You'll have to experiment to see what works best for you. If you eat before you train, ideally give yourself at least an hour to digest your food before you begin.

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[97] C. Roth et al., "Lean Mass Sparing in Resistance-Trained Athletes During Caloric Restriction: The Role of Resistance Training Volume," European Journal of Applied Physiology, February 11, 2022, https://doi.org/10.1007/s00421-022-04896-5.

After resistance training, your body is primed to convert any food you consume into energy to burn at a very high rate, so it can replace glycogen you used during the workout.[98] That does not mean you should gorge on pastries or pizza after training. Unless you're deliberately low-carb or keto, the meal following a resistance training workout ideally should include both natural sugars (a few pieces of fruit) and starchy complex carbohydrates (such as rice, potatoes, bread, pasta, or oatmeal) to replenish glycogen stores in your liver and muscles. It should also include some quality protein (such as whey, beef, chicken, fish, turkey, or eggs).

This meal doesn't have to be complicated. The enzymes that facilitate restocking of glycogen are upregulated after resistance training,[99] and you direct more of the amino acids coming from the protein you eat into your muscle tissue after resistance training too,[100] so it's a good idea to eat within

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[98] R. Pratley et al., "Strength Training Increases Resting Metabolic Rate and Norepinephrine Levels in Healthy 50- to 65-Yr-Old Men," Journal of Applied Physiology 76, no. 1 (January 1994): 133–37, https://doi.org/10.1152/jappl.1994.76.1.133.

[99] Yasuko Manabe et al., "Exercise Training-Induced Adaptations Associated with Increases in Skeletal Muscle Glycogen Content," FEBS Journal 280, no. 3 (February 2013): 916–26, https://doi.org/10.1111/febs.12085.

[100] Colin D. Wilborn and Darryn S. Willoughby, "The Role of Dietary Protein Intake and Resistance Training on Myosin Heavy Chain Expression," Journal of the International Society of Sports Nutrition 1, no. 2 (December 2004), https://doi.org/10.1186/1550-2783-1-2-27.

the three hours after training.[101] BUT, contrary to the lore in the 1990s, if you don't eat in this window, it's not the end of the world! As long as you don't eat far too few calories, you'll still make progress. You'll likely recover faster if you do eat in this window, but follow your hunger cues. There is a lot more to say about nutrition and training. Far more than can be covered here, but I provide a few more tips below, and even more in chapter 4.

## Hydration

Drink water before, during, and after a workout, especially on hot days or if you tend to sweat a lot. I cover hydration later in chapter 4 as well. Yes, there is a protocol for how to

[101] Erik A. Richter and Mark Hargreaves, "Exercise, GLUT4, and Skeletal Muscle Glucose Uptake," Physiological Reviews 93, no. 3 (2013): 993–1017, https://doi.org/10.1152/physrev.00038.2012; Erik A. Richter, "Is GLUT4 Translocation the Answer to Exercise-Stimulated Muscle Glucose Uptake?," American Journal of Physiology 320, no. 3 (February 2021): E240–43, https://doi.org/10.1152/ajpendo.00503.2020; Kendra D. Sticka et al., "Exercise Increases Glucose Transporter-4 Levels on Peripheral Blood Mononuclear Cells," Medicine & Science in Sports & Exercise 50, no. 5 (May 2019): 938–44, https://doi.org/10.1249/MSS.0000000000001528; John L. Ivy, "Muscle Glycogen Synthesis Before and After Exercise," Sports Medicine 11, no. 1 (1991): 6–19, https://doi.org/10.2165/00007256-199111010-00002; Jorn Trommelen et al., "The Anabolic Response to Protein Ingestion During Recovery from Exercise Has No Upper Limit in Magnitude and Duration In Vivo in Humans," Cell Reports Medicine 4, no. 12 (December 2023): 101324, https://doi.org/10.1016/j.xcrm.2023.101324; Andrew M. Holwerda, "Dose-Dependent Increases in Whole-Body Net Protein Balance and Dietary Protein-Derived Amino Acid Incorporation into Myofibrillar Protein During Recovery from Resistance Exercise in Older Men," Journal of Nutrition 149, no. 2 (February 2019): 221–30, https://doi.org/10.1093/jn/nxy263.

hydrate properly, as this is one of the most undervalued tools for mental and physical performance.

## Supplements

You do not need supplements to make progress with your resistance training. What's most important is that you train hard, get enough sleep, and eat right. If you want to explore supplements, however, there are two that have been shown to increase strength and hypertrophy: creatine monohydrate and whey protein.

Creatine brings more water into your muscles, enhancing training capacity and recovery ability, which thus makes you stronger. It also can have some positive effects on brain function,[102] especially after mild sleep deprivation.[103] I take five to ten grams of creatine monohydrate daily, as a powder dissolved in water. You can take it any time of day, with or without food. Any reputable brand with third-party testing and that is "Certified for Sport" is fine. I have seen no

[102] Chen Xu et al., "The Effects of Creatine Supplementation on Cognitive Function in Adults: A Systematic Review and Meta-Analysis," Frontiers in Nutrition 11 (July 12, 2024): 1424972, https://doi.org/10.3389/fnut.2024.1424972; Chen Xu et al., "Corrigendum: The Effects of Creatine Supplementation," Frontiers in Nutrition 12 (February 17, 2025): 1570800, https://doi.org/10.3389/fnut.2025.1570800.

[103] Ali Gordji-Nejad et al., "Single Dose Creatine Improves Cognitive Performance and Induces Changes in Cerebral High Energy Phosphates During Sleep Deprivation," Scientific Reports 14, article no. 4937 (2024), https://doi.org/10.1038/s41598-024-54249-9.

evidence that creatine harms your kidneys[104] or makes your hair fall out,[105] which is what some online claims say. It's normal to gain a few pounds of water-in-the-muscles weight with regular creatine use. If you stop taking it, you urinate out that extra fluid.

Whey protein is simply a high-quality, easily digested protein. It's convenient and cost-effective when compared to other quality proteins. It doesn't have many other micronutrients like meat or eggs do, however, so I wouldn't make whey protein your only quality protein source. I drink one to two scoops of whey protein mixed with water once or twice a day, which equates to 20 to 80 grams of protein (it's about 20 g of protein per scoop). Along with the foods I eat, this helps me meet my daily protein goal. Without whey protein I find it challenging to eat about 1 gram per pound of lean body weight each day.

The other supplement worth mentioning in the context of resistance training is caffeine, which has been shown to improve exercise performance when 100 to 400 milligrams is consumed about thirty minutes prior to and even during

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[104] Elham Kabiri Naeini et al., "Effect of Creatine Supplementation on Kidney Function: A Systematic Review and Meta-Analysis," BMC Nephrology 26, article no. 622 (2025), https://doi.org/10.1186/s12882-025-04558-6.

[105] Mohammadyasin Lak et al., "Does Creatine Cause Hair Loss? A 12-Week Randomized Controlled Trial," Journal of the International Society of Sports Nutrition 22, no. S1 (2025): 2495229, https://doi.org/10.1080/15502783.2025.2495229.

exercise.[106] That's about one to four cups of coffee, depending on the brew strength and size of the serving. Not everyone can or should consume caffeine, however. It makes some people anxious. I personally love caffeine and tolerate it very well, so I drink a cup or two of black coffee and/or a yerba maté about thirty minutes before resistance workouts.

A lot of people are into energy drinks these days as pre- and intra-workout energy sources. I will occasionally drink an energy drink (Jocko Go or Gorilla Mind—I know, wild names!—are the ones I like best for taste and effect) before and during a workout, but not every time. If I am facing a particularly challenging leg workout I tend to make my own pre-workout by drinking (or taking) 200 milligrams of caffeine and alpha-GPC (300–900 mg). I discuss alpha-GPC and other compounds more in chapter 6. Just remember, if you're working out in the evening, drinking caffeine beforehand will really disrupt your sleep.

There are a few other supplements which have been shown to improve exercise performance—like Rhodiola rosea,

[106] Nanci S. Guest et al., "International Society of Sports Nutrition Position Stand: Caffeine and Exercise Performance," Journal of the International Society of Sports Nutrition 18, no. 1 (2021), https://doi.org/10.1186/s12970-020-00383-4.

which can reduce fatigue during training,[107] but I consider it a distant fourth place to the three mentioned already.

But I want to reiterate: You do not need supplements to make progress resistance training. Focus on training properly! Then focus on recovery, meaning letting the adaptations you stimulated in the gym actually occur by getting proper sleep and nutrition. That is what really moves the needle.

## A week off every twelve to sixteen (or even ten to twelve) weeks

I attribute taking a week off from resistance training every three to four months or so as one of the main reasons I have been able to make regular progress in resistance training over the last thirty-five years. I know that sounds crazy, and that it runs counter to much of the "no days off" advice seen online, but I am convinced it's the way to go, for three reasons.

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[107] Zhaolong Wang et al., "Effects of the Combined Supplementation of Caffeine and Rhodiola rosea with Resistance Training on Lower Limb Explosive Power in Male Volleyball Players," Nutrients 17, no. 4 (2025): 681, https://doi.org/10.3390/nu17040681; Xiaolin Wang et al., "The Effect of Rhodiola rosea Supplementation on Endurance Performance and Related Biomarkers: A Systematic Review and Meta-Analysis," Frontiers in Nutrition 12 (September 25, 2025): 1645346, https://doi.org/10.3389/fnut.2025.1645346; Grant M. Tinsley et al., "Rhodiola rosea as an Adaptogen to Enhance Exercise Performance: A Review of the Literature," British Journal of Nutrition 131, no. 3 (August 2023): 461–73, https://doi.org/10.1017/s0007114523001988.

First, it gives your joints and your mind a chance to reset. Any nagging aches or residual muscle soreness tend to go away during that week off, and you come back feeling reset.

Second, most people make actual progress during the week off. Training-induced stress is cumulative; when we train and recover between workouts, we get stronger and can manage more reps and/or more weight in perfect form from week to week, but not all our bodily tissues adapt at the same rate. By the time we are eight, nine, or even fourteen weeks into training since our last week off, we tend to have a harder time recovering from workouts. Taking a week off lets everything catch up, and lets you come back stronger.

Third, it's common that after twelve to sixteen weeks of training we feel less motivated to keep training hard. The relatively low frequency of resistance training I recommend in this program—just three times per week—usually staves off boredom but some people do start to lose steam after ten or more weeks of training. If you find yourself dragging yourself to the gym, and you're more interested in texting than training once you're there, take a week off, and come back. That usually does the trick. You'll be more energetic when you do.

Keep in mind, too, that sometimes we are just forced to take a week off because life says we have to! I don't think training when you're sick is a good idea, and it gets others

sick, too, so please stay out of the gym until you're better! Other times we are forced to take a week off because we have a major deadline or a family vacation or other demands. I personally like to train on vacations because it helps me get over jet lag when I travel, but you may feel like it's a waste of time. It's a matter of choice, and how you view vacation.

Regardless of whether your week off is due to a cold or pre-planned for the end of a twelve to sixteen week stretch of training, or simply because a work deadline hits, don't worry about it causing a setback. It's part of the process. Just get back to training (properly) after that week off.

## COMMON QUESTIONS

**Does it matter what day I begin this sequence**

No. Start it on Sunday or Tuesday or whenever you like. What's important is to follow the sequence.

**You listed abdominals/core as "required" but neck and forearms as optional. Why? And am I supposed to train neck and abdominals/core and forearms the same way I train other muscles, with progressive overload, to failure, et cetera**

Some people need to work on their forearms—and their grip strength—and others don't, for a variety of reasons. It could be genetics, or because that second group targets their forearms and grip strength with plenty of work from

deadlifts, pull-ups, rows, and similar movements. Grip strength is a correlate of longevity but not a cause. It's also one of the things that starts to wane as we age. I like training forearms when I train shoulders and arms, and just for three to four sets of flexor and/or extensor movements (see protocolsbook.com/videos for examples). Use higher rep ranges, say twelve to thirty reps—no need to train forearms with lower reps and heavy weight. You'll get that indirectly from your heavier pulling movements when you train your back, such as chin-ups or rows.

Abdominals are a specific set of muscles in your midsection and your "core" includes a lot more than just these. I suggest you train your abs and core one to two times per week. Endless crunches are not the way to go. There are better options to train these muscles. And some let you train your forearms and strengthen your grip, and let you get in a great stretch too, all at once. Like hanging from a bar and doing knee raises, then progressing over time to hanging chair sits and L-sits, where you hold your legs straight out, parallel to the floor, and eventually hanging pikes, where you bring your shins up to the bar. You don't just leap into pikes, by the way; you work toward these with the exercises above, and everyone—yes, everyone (men and women)—I have taken through this progression has eventually been able to do multiple sets of three to five reps of hanging pikes! They were surprised, but I was not. Your body is amazing in how

it will adapt if you give it the right stimulus, and the right recovery. (Oh, and I know three to five reps is not ten, but when you reach that number you will have a very strong core and grip!)

Other reasons to train your core: It's vital for posture and back health, and it will help your resistance and other fitness training. Six-pack abs are the consequence of having low enough body fat to see the contour of the abdominals. You don't get that from training alone. But abs and other muscles of your core react to progressive overload just like any other muscle (here too, see protocolsbook.com/videos for examples).

Neck exercises are the least performed of any exercises in any gym, but they are so important! Your neck is your upper spine, and a strong neck is crucial for posture. It also helps to keep your shoulders uninjured. Also, done right they won't give you a huge neck—or (oxymoronically) the "no-neck" look. Women and men who train their neck properly can avoid that look if they want to, and yet can still enjoy the better posture and facial tone that work can bring, as the neck and face muscles both activate during these exercises. For wide-shouldered men, balanced neck development is crucial to avoid appearing as if your head is too small for your body! This will be of great aesthetic and structural benefit. Neck training is simple to do safely and effectively

once you know how, and requires little or no equipment. I include several examples also at protocolsbook.com/videos.

Everyone should be sure to train their rear deltoid muscles—small muscles on the back of the shoulders that, when made sufficiently strong, greatly improve posture and shoulder stability. There is a way to train this muscle along with other muscles of the middle back that will also help make your default posture more upright. The exercise is called a “face pull,” and no, it does not involve attaching weights to your face or jaw. You do them during your “shoulders and arms” resistance training day, using exercise bands or a cable machine. I have a video of the person who popularized these, Jeff Cavaliere from AthleanX, teaching me the proper form at protocolsbook.com/videos.

**What if something comes up and I have to miss a workout, what then**

I address this in Exercise Protocol 3, because the resistance training outlined here is part of a larger program. In the meantime, know that you have a fair degree of flexibility, and you can move a workout forward or back a day if needed.

**Can I still play sports and do other activities**

Of course. You’ll be stronger for it. And the relatively low volume of sets in this program means you won’t be sore all

the time. (Again, soreness is zero indication of an effective workout—it does not mean you effectively stimulated specific hypertrophy or strength adaptations.) Please don't add additional resistance training work or CrossFit workouts and the like until you've tried this program, but tennis, Pilates, hiking, yoga, archery, golf, soccer, et cetera, and being active with house and yard work is encouraged. Most people nowadays are too sedentary. More movement is nearly always better than less. And of course, cardiovascular training is key. I discuss that next.




EXERCISE PROTOCOL

## Do Cardiovascular Training Three Times per Week

### WHAT TO DO

- Elevate your heart rate for anywhere from six to seventy-five minutes, three times per week. Plan three sessions, each focused on stimulating a different specific adaptation: a long, a medium, and a brief cardiovascular workout, scaling the intensity inversely with the duration of each session. That means the long cardio workout should be of low intensity, the brief cardio workout should be of high intensity. Here is how to incorporate cardio into the overall program:

- Day 1: Cardiovascular training, 45–75 minutes (low intensity/long duration)
- Day 2: Resistance training (legs: calves, quadriceps, hamstrings, glutes)
- Day 3: Rest day
- Day 4: Resistance training (torso: chest, back, abdominals/core; optional neck)
- Day 5: Cardiovascular training, 25–30 minutes (moderate intensity/moderate duration)
- Day 6: Cardiovascular training, 6–17 minutes (high intensity/brief duration)
- Day 7: Resistance training (shoulders, biceps, triceps, abdominals/core; optional forearms and/or neck)

The first cardio workout each week marks the first day of the foundational fitness program schedule. It should be relatively low intensity, and long duration. And it should be continuous. Your goal is to sustain a steady output of effort for forty-five to seventy-five minutes without any pauses except perhaps to drink some water or wait for a stoplight to change so you can cross the street. This falls into the category of classic endurance work. The primary adaptation you're stimulating with this workout is to increase your heart's stroke volume, which is the amount of blood your heart pumps with each beat. This means the heart can pump

more blood per beat, and thus does not have to beat as fast to deliver the same amount of oxygen. The consequences of that for your health are manifold, including a lower resting heart rate when not exercising.[108] This workout also stimulates a number of metabolic pathways related to utilizing fat for fuel and for vascular growth and even lymphatic health.[109] It indirectly improves mood and sleep too.[110]

The challenge of this workout resides in its duration, not its intensity, and for many people the most challenging aspect is refraining from going faster and harder as they complete it. The forty-five to seventy-five minute duration is

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[108] Ilkka Heinonen, "Cardiac Output Limits Maximal Oxygen Consumption, but What Limits Maximal Cardiac Output?," Experimental Physiology 110, no. 5 (2025): 666–74, https://doi.org/10.1113/EP091594.

[109] Kyle Fulghum and Bradford G. Hill, "Metabolic Mechanisms of Exercise-Induced Cardiac Remodeling," Frontiers in Cardiovascular Medicine 5 (September 10, 2018): 127, https://doi.org/10.3389/fcvm.2018.00127; Huihua Chen et al., "Exercise Training Maintains Cardiovascular Health: Signaling Pathways Involved and Potential Therapeutics," Signal Transduction and Targeted Therapy 7, article no. 306 (2022), https://doi.org/10.1038/s41392-022-01153-1; Kansan Roh et al., "Exercise-Induced Cardiac Lymphatic Remodeling Mitigates Inflammation in the Aging Heart," Aging Cell 24, no. 6 (June 2025): e70043, https://doi.org/10.1111/acel.70043.

[110] Majd A. Alnawwar et al., "The Effect of Physical Activity on Sleep Quality and Sleep Disorder: A Systematic Review," Cureus 15, no. 8 (2023): e43595, https://doi.org/10.7759/cureus.43595; Weicheng Li et al., "Influence of Aerobic Exercise on Depression in Young People: A Meta-Analysis," BMC Psychiatry 24, article no. 571 (August 2024), https://doi.org/10.1186/s12888-024-06013-6; James A. Blumenthal et al., "Exercise and Pharmacotherapy in the Treatment of Major Depressive Disorder," Psychosomatic Medicine 69, no. 7 (2007): 587–96, https://doi.org/10.1097/PSY.0b013e318148c19a.

intentionally broad. In fact, if you're out of shape when starting this program, I suggest you begin with just fifteen to twenty minutes, and add a few minutes to that each week, until you're doing forty-five to seventy-five minutes of continuous movement. In my observation, most people find that sixty minutes is about the upper limit of their psychological tolerance for this sort of exercise, unless they really love the activity or it's made social, like a hike or bike ride with family or friends. In any case, your goal pace throughout should be such that you can just barely hold a conversation. It's quite all right if you must pause your speech here and there to catch your breath, but the idea is to exert yourself at a level that, were you to push yourself any faster or harder, holding a conversation would be impossible.

Another way to calibrate to that goal pace is to push as hard as you can while still breathing in and out only through your nose. No mouth breathing whatsoever, unless you are talking. As soon as you feel you have to gasp to get air through your mouth, you know you're pushing too hard.

In the language of percentage of maximum heart rate this cardio session is done at 60 to 70 percent of your maximum heart rate (HR), but ideally closer to 70 percent. Your max HR can be estimated by this formula: 220 – your age = your approximate max HR. That's not a perfect measurement but it's close, and for most it's more than sufficient to use here.

If you want to be more precise about it, max HR is better approximated by the following formula: for men, you take 208 beats per minute – (0.7 × your age in years); and for women, it's 206 beats per minute – (0.88 × your age in years).[111] Once you have determined your approximate max HR, multiply that by 0.6 and 0.7 to get the lower and upper range, respectively, of your target heart rate for this cardio session. I've shared this formula because some people like to use an HR monitor while training to be sure they stay in that target HR zone. I prefer to use the "upper limit of effort while maintaining nasal breathing" approach, because whenever I have measured my HR while running and restricting myself to nasal breathing, it lands at about 70 percent (+/–5 percent) of my max HR. I have nothing against HR monitors. I just prefer not to use one when doing this workout. I like to let my mind wander or to listen to a book or podcast or music but if you like tracking your HR in real time, go for it.

A few other things to consider for this workout: Some people refer to this as Zone 2 cardio, which is a term that describes the intensity of the exercise with regard to heart rate training zones. But on that scale, I consider it more Zone

[111] Hirofumi Tanaka et al., "Age-Predicted Maximal Heart Rate Revisited," Journal of the American College of Cardiology 37, no. 1 (2001): 153–56, https://doi.org/10.1016/s0735-1097(00)01054-8; Martha Gulati et al., "Heart Rate Response to Exercise Stress Testing in Asymptomatic Women: The St. James Women Take Heart Project," Circulation 122, no. 2 (2010): 130–37, https://doi.org/10.1161/CIRCULATIONAHA.110.939249.

⅔ because it's done at the high end of Zone 2 and the low end of Zone 3.[112]

Ideally you would get outside to do this workout so that you're getting all the benefits that come with being outdoors. This raises the issue of what type of exercise is best for this. That's an easy one to answer. Select any activity you can do for the specified duration and intensity without getting hurt. That means do what is safe for your joints, works for your skill level, doesn't stress any of your injuries, and can be completed in safe conditions. I love running and much prefer doing it outside on a trail or city streets to all other forms of cardio, but many people loathe running, or it hurts their knees, or it's not safe for them to run in a given area. You don't have to run! You can opt for cycling, or use the elliptical machine or a stair climber, or hike with a weight vest or a rucksack or, heck, even a backpack with a one-gallon jug of water in it. The possibilities are nearly infinite, and many of these cost very little. If I don't run, I'll sometimes swim laps in a pool or swim parallel to shore in the ocean. If the weather is bad, I'll occasionally run on a treadmill or use a stationary bike.

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[112] Øystein Sylta et al., "From Heart-Rate Data to Training Quantification: A Comparison of 3 Methods of Training-Intensity Analysis," International Journal of Sports Physiology and Performance 9, no. 1 (2014): 100–107, https://doi.org/10.1123/IJSPP.2013-0298.

Pick your poison and deliver it slow and steady for forty-five to seventy-five minutes. Many people find this workout meditative and learn to enjoy it. While you can of course go longer than seventy-five minutes, remember you're training your legs the next day.

The second cardio workout each week is done three days after your leg workout (day five of the program). By then, your legs should be feeling mostly or completely recovered. This second cardio workout should be moderately fast-paced. It should have you breathing fairly hard but not gasping for air or needing to stop to recover. I suggest you start with five minutes of low-intensity movement to warm up and then shift to twenty-five to thirty minutes of "work" where your heart rate is kept between about 75 and 85 percent of your maximum heart rate.

Using the heart rate training zones, this workout falls in the high Zone 3 and low Zone 4 range.

Notably, although this workout is more intense than the first cardio session, it generally leaves you feeling energized and focused for many hours afterward. The key is to not push overly hard at any one time of that approximately twenty-five to thirty minutes. You want to stay roughly in that "75 to 85 percent of max HR" range. That said, it's perfectly all right to move between high and low within this range by pushing harder at times, and pulling back at times.

Say, for example, you're climbing a hill or you're just feeling strong on your bike for a stretch and want to push yourself. Your pace and your heart rate don't have to be perfectly constant throughout. Remember: It's not a sprint! You don't want to overly tax your body. The third cardio workout is the one that is intense. Luckily, it's brief too.

The third cardio workout each week lands four days following your leg workout. If you still have a bit of residual leg soreness, no big deal: Proceed anyway. Some people refer to this type of workout as Zone 5 training or HIIT (high-intensity interval training), but I like to refer to it as "max heart rate training" or simply "intense cardio" because there are non-interval ways to do it, and—for reasons that will become clear in a minute—getting your heart pumping near your maximum heart rate is highly beneficial.

There are a few variations in the "classic" versions of this workout. All begin with five to ten minutes of warming up at an easy to moderate pace. HIIT has you modulating between near-all-out efforts that put you at 90 to 95 percent of your max HR, interspersed with bouts of lower intensity effort where you let your HR fall a bit before repeating the sequence.[113] Another approach is to start at an easy effort

[113] Andrea D'Amuri et al., "Effectiveness of High-Intensity Interval Training for Weight Loss in Adults with Obesity: A Randomised Controlled Non-Inferiority Trial," BMJ Open Sport & Exercise Medicine 7, no. 3 (2021): e001021, https://doi.org/10.1136/bmjsem-2020-001021.

and build in a continuous and progressive manner—for example, increasing the level of resistance to your effort (think: increasing treadmill speed and incline)—until your effort plateaus, or starts to drop. (That actually mirrors how a VO2 max test, a measure of fitness and endurance, is conducted.) There are many ways to do max heart rate training but the total time spent at near-all-out effort is always pretty brief: as short as about six minutes, and up to about sixteen minutes. Either works for this third cardio session.

What form of exercise you choose can vary. For example, you could hop on an assault bike—those stationary bikes with a fan and handles that you pump forward and back while you pedal. You'll soon realize that the fan is there to increase the resistance, not just to cool you down. Or you could use a stationary bike that lets you adjust the resistance of the pedals. You could also go sprint at a track or on a trail, or swim in a pool. The important thing is to do something in which you can really exert yourself without getting injured.

A "fun" HIIT session is this: brief five- to ten-minute warm-up, then pedal (or run, row, or swim, etc.) as hard and fast as you can for thirty seconds, then rest or slow down for sixty seconds, then another thirty seconds of all-out effort, then back again to sixty seconds of rest or low effort, and so

on, for a total of six to twelve rounds of alternations. I do this on the assault bike once a week. I love-hate it.

You will quickly realize that with each round of all-out effort it gets tougher and tougher to recover, such that by rounds six to twelve, those sixty seconds of rest will seem to go by extremely fast. There is actually a neuroscientific basis for the warping of time perception we experience under stress that I discuss in the next chapter. The good news is, even though this workout is intense, the whole thing, including the warm-up, only takes about twenty-five minutes. At the end, you will feel fatigued, but remarkably, within five to ten minutes after stopping you will likely feel extra energized from all the adrenaline and norepinephrine you generated in your body and brain.[114] You will also have provided a great stimulus to improve your VO2 max, and max HR, the importance of which I cover below.

Other options for this workout include the Cooper test, where you see how far you can go (run, swim, cycle, etc.) in twelve minutes of steady, hard effort.[115] That is a “fun” one too.

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[114] Stephen H. Boutcher, “High-Intensity Intermittent Exercise and Fat Loss,” Journal of Obesity 2011 (November 2010): 868305, https://doi.org/10.1155/2011/868305.

[115] Kenneth H. Cooper, “A Means of Assessing Maximal Oxygen Intake: Correlation Between Field and Treadmill Testing,” JAMA 203, no. 3 (1968): 201–4, https://doi.org/10.1001/jama.1968.03140030033008.

Or you can try to run your best mile time. This is something performance coach, author, and former elite runner Steve Magness has emphasized as being a terrific correlate of VO2 max.[116] It's one of my favorite workouts. Here is how to do it properly:

First find a 400-meter track. Start by warming up with an easy jog of four to six laps slowly interspersing the occasional faster-paced 100 meters—not all out but just quickening your pace to about 60 to 70 percent of your subjective "run for your life" speed. Perhaps do that on one straight-track segment between turns on each warm-up lap. Once you feel properly warmed up, run four laps (which is 1,600 meters—about 9 meters shy of 1 mile). It's a skill to run the mile and pace everything just right for your best time. But you're not aiming to be a competitive miler here. So set out at a pace you can maintain for three laps—about 85 percent of "running for your life" speed, then push it harder on the last lap, or 95 percent of "running for your life" speed. The key is to pace yourself so you're not sprinting laps one to three and then forced to walk or jog the remainder. It can take a bit of practice but within a few weeks you'll know the pace to take for each lap. Almost everyone likes to really push the final 200 meters or so.

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[116] Steve Magness, "Longevity and Vo2max? Does It Actually Matter?," Steve Magness Substack, November 10, 2025, https://stevemagness.substack.com/p/longevity-and-vo2max-does-it-actually.

I test my approximate best mile time about once every two months lately. I'm not training to be a competitive miler but it's nice to have one version of the short intense cardio workout where I can track my progress across the year. I doubt I'll beat my best time from high school ever again, but it's fun to have an "a bit faster than last month" run. My goal is to maintain or reduce my mile time each time I run it.

And there is the Norwegian 4×4, if you want to try it.[117] That's where you warm up, and then run, bike, or row hard for four minutes, then rest or slow down for four minutes. Repeat this cycle for a total of four work-rest rounds.

As you can see with all these examples, the exact work-to-rest ratio and total time can vary dramatically, and yet all can be effective. What's key is that during the work periods, you're truly working hard. That is because, during the rest periods, you're allowing enough time for recovery so that you can again come back to a truly hard effort. Don't be overly caught up in the exact protocols. You can alter them if you like and still get the same results.

I'm recommending three different types of cardio training sessions every week because this variation is typically what delivers the best results over time for improving your VO2 max. We're trying to balance duration and intensity in this

[117] Dorthe Stensvold et al., "Effect of Exercise Training for Five Years on All Cause Mortality in Older Adults—the Generation 100 Study: Randomised Controlled Trial," BMJ 371 (2020): m3485, https://doi.org/10.1136/bmj.m3485.

program. It's best to emphasize both. You want longer moderate efforts and you want brief and very intense efforts. And while the shorter, intense efforts are extremely effective (and time efficient), they take longer to recover from compared to the longer, moderate efforts. Given that you're also balancing this cardio training with the resistance training, a blend of longer/lower, moderate, and brief/high intensity cardio training sessions is what works best for results across the board.

## HOW IT WORKS

The cardio workouts described above each help trigger beneficial adaptations. They all increase your heart's efficiency in delivering blood (and thus oxygen and vital nutrients) to the rest of the body.[118] They trigger capillary growth in your muscles, which allows more oxygen to move through these vessels during all activities.[119] They stimulate an increase in the number, size, and efficiency of your mitochondria, the energy producing organelles in our cells known to be critical for preventing chronic disease and premature aging. Further, they all acutely elevate levels of

[118] Ole Johan Kemi and Ulrik Wisløff, "High-Intensity Aerobic Exercise Training Improves the Heart in Health and Disease," Journal of Cardiopulmonary Rehabilitation and Prevention 30, no. 1 (January 2010): 2–11, https://doi.org/10.1097/hcr.0b013e3181c56b89.

[119] Knut Sindre Mølmen et al., "Effects of Exercise Training on Mitochondrial and Capillary Growth in Human Skeletal Muscle: A Systematic Review and Meta-Regression," Sports Medicine 55, no. 1 (October 2024), https://doi.org/10.1007/s40279-024-02120-2.

the catecholamines: dopamine, epinephrine, and norepinephrine.[120] Also, because they rely on a mixture of aerobic (oxygen dependent) and anaerobic (oxygen independent) processes, cardio of the types in this program increases metabolic flexibility, or your body's ability to rely on different fuel sources for energy. And that's just a small sampling of the many beneficial adaptations.[121]

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[120] Neil L. Coplan et al., "Exercise-Related Changes in Serum Catecholamines and Potassium: Effect of Sustained Exercise Above and Below Lactate Threshold," American Heart Journal 117, no. 5 (May 1989): 1070–75, https://doi.org/10.1016/0002-8703(89)90864-8.

[121] Zoladz, Muscle and Exercise Physiology; Powers and Howley, Exercise Physiology; Koman et al., "Beneficial Effects of Aerobic Exercise"; MoTrPAC Study Group et al., "Multi-Omic, Multi-Tissue Responses"; Contrepois et al., "Molecular Choreography of Acute Exercise"; Kraemer and Ratamess, "Hormonal Responses and Adaptations"; Zouhal et al., "Catecholamines and the Effects"; Athanasiou et al., "Endocrine Responses of the Stress System"; Hellsten, "Cardiovascular Adaptations to Exercise Training"; Mark Ross et al., "Exercise-Induced Skeletal Muscle Angiogenesis," European Journal of Applied Physiology 123, no. 7 (July 2023): 1415–32, https://doi.org/10.1007/s00421-022-05128-6; David A. Hood et al., "Exercise and the Regulation of Mitochondrial Turnover," Progress in Molecular Biology and Translational Science 135 (2015): 99–127, https://doi.org/10.1016/bs.pmbts.2015.07.007; Tao Cai et al., "The Role of Exercise-Mediated Mitochondrial Quality Control Remodeling in Aging," Frontiers in Cell and Developmental Biology 14 (March 2026): 1792645, https://doi.org/10.3389/fcell.2026.1792645; Julia C. Basso and Wendy A. Suzuki, "The Effects of Acute Exercise on Mood, Cognition, Neurophysiology, and Neurochemical Pathways," Brain Plasticity 2, no. 2 (March 2017): 127–52, https://doi.org/10.3233/BPL-160040; Bret H. Goodpaster and Lauren M. Sparks, "Metabolic Flexibility in Health and Disease," Cell Metabolism 25, no. 5 (May 2017): 1027–36; Mølmen et al., "Effects of Exercise Training on Mitochondrial"; Joanna Kruk et al., "Physical Exercise and Catecholamines Response: Benefits and Health Risk: Possible Mechanisms," Free Radical Research 54, no. 2–3 (February 2020): 105–25, https://doi.org/10.1080/10715762.2020.1726343.

The extent to which brief, high-intensity or longer, lower-intensity cardio triggers these different adaptations varies, but there is also synergy between them.[122] Those brief, intense cardio training sessions will allow you to move faster but with less effort on your long and medium cardio days, not unlike how getting stronger with low repetition heavy sets in the gym equates to you being able to move more weight on your higher repetition sets once you get stronger.

So, while people often debate which form of cardiovascular exercise is best for overall health and longevity, that's not the lens through which to look at it. I suggest a practical lens. (And you know by now, I don't believe in any best type of exercise; only the best way to structure an effective exercise program.) For instance, consider that your one-mile time is a good reflection of your VO2 max.[123] Does that mean you should only do one mile distances for cardio? No. In fact, whether you're a competitive miler or simply using the one-mile time test as a fun workout and barometer of your VO2 max, doing some longer/lower, moderate, and

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[122] David P. Swain and Barry A. Franklin, "Comparison of Cardioprotective Benefits of Vigorous Versus Moderate Intensity Aerobic Exercise," American Journal of Cardiology 97, no. 1 (January 2006): 141–47, https://doi.org/10.1016/j.amjcard.2005.07.130.

[123] Laura Weiglein et al., "The 1-Mile Walk Test Is a Valid Predictor of VO2max and Is a Reliable Alternative Fitness Test to the 1.5-Mile Run in U.S. Air Force Males," Military Medicine 176, no. 6 (June 2011): 669–73, https://doi.org/10.7205/milmed-d-10-00444.

brief/high intensity cardio workouts across the week is how to get faster at the mile.

In science we love to isolate variables, so people have done the "which cardio exercise is best?" experiment. That research made clear that if you have to choose one type, you want to opt for brief/high intensity cardio rather than, say, three thirty- to forty-five-minute Zone ⅔ sessions.[124] But doing a mix of durations and intensities is recommended, and for the reasons mentioned above, it meshes well with your resistance training. A Zone ⅔ jog or hike for an hour the day before you train your legs with weights is no problem. Sprinting the day after leg training to failure? Not a good idea. The injury risk is too high.

Doing these three cardio training sessions also means you're logging a total of about two hours per week of cardio. Considering the immediate and long-term health benefits, that's a great return on investment.

We hear all the time about "fitness" and it reflects many things. But from a health perspective consider that two studies—one published in 2018 and another in 2022—both showed essentially the same thing, which is that as you move from the lowest to the higher tiers of cardiorespiratory

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[124] Sverre Valstad et al., "Comparison of Long and Short High-Intensity Interval Exercise Bouts on Running Performance, Physiological and Perceptual Responses," Sports Medicine International Open 2, no. 01 (December 2017): E20–27, https://doi.org/10.1055/s-0043-124429.

fitness (one measure of that is your VO2 max), your mortality risk drops.[125] These are meaningful shifts in life expectancy too. One of the studies states, “Those with the highest fitness had roughly five times lower mortality than the least fit.”

Resistance training is less about cardiorespiratory fitness, but its effects in maintaining (or increasing) muscle mass and bone density, promoting cognitive function and healthy body recomposition, make a combination of resistance training and cardio training two of the three vital interlocking puzzle pieces for excellent physical health.

## COMMON QUESTIONS

### Does doing three cardio training sessions per week prevent you from building muscle? And does resistance training harm my cardio efforts

To be clear: If your goal is purely to put on as much muscle as possible, or to do an endurance event like a half marathon or marathon as fast as possible, there are better training programs for those specific goals. When splitting your time across both building muscle and building endurance, there’s

[125] Kyle Mandsager et al., “Association of Cardiorespiratory Fitness with Long-Term Mortality Among Adults Undergoing Exercise Treadmill Testing,” JAMA Network Open 1, no. 6 (October 2018): e183605, https://doi.org/10.1001/jamanetworkopen.2018.3605; Prathiyankara Shailendra et al., “Resistance Training and Mortality Risk: A Systematic Review and Meta-Analysis,” American Journal of Preventive Medicine 63, no. 2 (May 2022): 277–85, https://doi.org/10.1016/j.amepre.2022.03.020.

going to be a performance trade-off. You can't maximize for both at the same time because of so-called interference effects.[126]

That said, for vitality, health, longevity, aesthetics, and all-around fitness, you want to do both resistance training and cardio training. Research indicates that the interference effects are minimal when done on separate days at the duration and intensities similar to those in this foundational program.[127]

### What if I play a sport

If you're training for a competition or a race, or you play a sport like soccer or tennis or basketball, and you play hard for an hour or more one to two times per week, the program here is still great. You may want to do the lower number of work sets, and you'll really need to tend to your recovery and nutrition by making sure you get enough hours of quality sleep and calories—including, yes, enough quality protein, but also starches and fats.

Again, the program here will get you the most all-around fit as possible at any age. That does not mean you won't gain

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[126] G. A. Dudley and R. Djamil, "Incompatibility of Endurance- and Strength-Training Modes of Exercise," Journal of Applied Physiology 59, no. 5 (November 1985): 1446–51, https://doi.org/10.1152/jappl.1985.59.5.1446.

[127] Tao Wang and Shumin Bo, "Optimizing Concurrent Training Programs: A Review on Factors That Enhance Muscle Strength," Medicine 103, no. 52 (December 2024): e41055, https://doi.org/10.1097/md.0000000000041055.

strength and muscle—you will. You will also burn fat. You'll increase your endurance too. And once you learn how different exercises impact your body, you can emphasize some and reduce others, per the suggestions I describe in this protocol. Exercise done correctly is very potent in that way. Assuming you pay attention to the points I make about recovery too—that is, eat well and sleep well—I'm certain you'll achieve remarkable results with this program.

### What if I haven't slept well, should I train or not

If you are healthy, and you happen to have had a bad night's sleep—by which I mean only one to two hours less sleep than usual—then try to stick with your workout. There is a growing amount of data that say it can help offset some of the short-term cognitive disturbances from sleep loss.[128]

Just be careful not to push too hard in the gym (or in life) after a poor night's sleep as injuries happen more often when people have not slept enough.[129] And don't make it a regular

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[128] Abdulmenaf Korkutata et al., "The Impact of Exercise on Sleep and Sleep Disorders," npj Biological Timing and Sleep 2, article no. 5 (2025), https://doi.org/10.1038/s44323-024-00018-w; Ali Gordji-Nejad et al., "Single Dose Creatine Improves Cognitive Performance and Induces Changes in Cerebral High Energy Phosphates During Sleep Deprivation," Scientific Reports 14, no. 1 (February 28, 2024): 4937, https://doi.org/10.1038/s41598-024-54249-9.

[129] Jonathan Charest and Michael A. Grandner, "Sleep and Athletic Performance: Impacts on Physical Performance, Mental Performance, Injury Risk and Recovery, and Mental Health," Sleep Medicine Clinics 15, no. 1 (2020): 41–57, https://doi.org/10.1016/j.jsmc.2019.11.005.

thing to get less than enough sleep, and expect to remedy that with exercise. You'll end up sick, injured, or both. Maybe do some non-sleep deep rest (NSDR) (see Sleep Protocol 8) and then push through the workout. If you have had a really bad night's sleep for two or more nights in a row, for whatever reason, I suggest you get back on track with your sleep before exercising again. Consider some deliberate heat exposure (sauna or a hot bath, for example), and if you feel you must exercise, try a restorative activity, like stretching, light yoga, or just walking.

**Is it better to eat before or after training**

There is no clear rule here. Personally, I prefer to train in the morning on an empty stomach. Figure out what works for you. It may be best to eat prior to training if you wake up hungry and you know that hunger will distract you during your workout, or you get dizzy or nauseous training on an empty stomach. Some people may find that eating about sixty to ninety minutes prior to exercise gives them an energy boost without the discomfort of feeling full. It's highly individual. Experiment with your timing of your workouts and meals.

### If I miss a day of cardio or resistance training, should I change the weekly schedule

It's hard to answer this one because… it depends. However, I would not change the order of your resistance or cardio training sessions across the week because the schedule starts to fall like dominoes, and it's hard to get back on track. But maybe something comes up and you can't do the long, slow cardio day as planned. I would not recommend moving that workout to the same day you train legs. I would simply skip the long cardio day and train your legs on schedule. Skipping leg day is a bad idea. You do three days of cardio per week, so having to miss one every now and again is not a huge deal. But you only train your legs directly once a week, and done correctly as outlined above it creates major positive systemic shifts in your brain and body. So when people say, "Don't skip leg day," it's not just about aesthetics. In fact, if I could only train once per week in the gym and do one cardio training session—not what I recommend, but if I had to—I would do leg day and I would do some sprints followed by light jogging four or five days later. That's just to illustrate a point about the potency of those two workouts. I recommend the full program described in this chapter.

Let's say you can't do the moderate cardio training session for whatever reason. You could, in theory, shift it to immediately after your arms workout that same day. But remember, you're scheduled to do the long, slow cardio

session the very next day, and you don't want your legs to feel like noodles when you wake up the following morning for leg day! So think ahead a bit about your recovery.

I suggest you stick to the schedule as best you can, missing days here and there or doubling up when there is no other option. Plan your workouts, follow through, and if people get curious about how you're getting such great results, teach them the principles, or even better, share the program. No one owns this or any exercise program; they are all derivatives of each other anyway, and based on what science says works.




EXERCISE PROTOCOL

## Incorporate Brief Mobility and Flexibility Work Two to Four Times per Week

### WHAT TO DO

- Do four specific movements to improve your mobility, two to four times a week.
- Refer to the instructions below to target areas susceptible to injury as you do these movements. Wrist, back, knee, hip, and shoulder injuries are common. The movements described here will ensure mobility in these areas. They aren't a guarantee against injuries, but they can help tremendously.

## HOW IT WORKS

First, I'm not giving you yet another workout to do. The current program—about three hours a week of resistance training, about two hours of cardio—is not terribly time-consuming, considering the years (literally!) it can add to your life. Still, I acknowledge that people's time is limited. That is why I suggest incorporating just a few minutes of mobility exercise two to four times per week. It can be done at the end of one of your other workouts, or any time of day really. It does not take long at all, only about ten minutes per session.

That might seem woefully inadequate to the mobility experts out there. I get it. I would love to add two thirty-minute mobility workouts to my own schedule but alas "my dance card is full," as they say. I'm assuming yours is too.

The good news is that some aspects of the training suggested in the earlier protocols in this chapter are pro-mobility and pro-flexibility too. Remember the emphasis on using a full range of motion for resistance training sets, as described in Exercise Protocol 2? Doing that will not only build your strength and muscles but can also enhance your flexibility and mobility. In fact, some studies find that full-range resistance training can be as effective as static stretching, a type of stretching in which you hold a muscle in a stretched, or elongated, position and gradually deepen the stretch using

body weight. Controlling the eccentric, or lowering, phase in full-range resistance training may be critical for this effect, which makes sense as that is when the muscle is elongating.[130]

There are easy ways to get even more flexibility from resistance training too. In the resistance training video tutorials at protocolsbook.com/videos, I also emphasize some exercises that are hardest for a given muscle when the muscle is in the elongated, stretched position, as well as exercises where the tension is greatest when the muscle is in its shortened, contracted position. Consider a stiff-legged deadlift or similar variation for your hamstrings, glutes, and lower back—the so-called posterior chain. It involves holding a barbell or dumbbells and moving from standing to something resembling a toe-touch (or an attempt to) by

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130 Shahab Alizadeh et al., "Resistance Training Induces Improvements in Range of Motion: A Systematic Review and Meta-Analysis," Sports Medicine 53, no. 3 (January 2023): 707–22, https://doi.org/10.1007/s40279-022-01804-x; Morton Rosenfeldt et al., "Comparison of Resistance Training vs Static Stretching on Flexibility and Maximal Strength in Healthy Physically Active Adults," BMC Sports Science, Medicine and Rehabilitation 16 (June 2024): 142, https://doi.org/10.1186/s13102-024-00934-1; Sebastian Vetter et al., "The Effects of Eccentric Strength Training on Flexibility and Strength in Healthy Samples and Laboratory Settings: A Systematic Review," Frontiers in Physiology 13 (April 2022): 873370, https://doi.org/10.3389/fphys.2022.873370; Francesco Favre et al., "Influence of Resistance Training on Joint Flexibility in Healthy Adults: A Systematic Review, Meta-Analysis, and Meta-Regression," Journal of Strength & Conditioning Research 39, no. 3 (December 2024): 386–97, https://doi.org/10.1519/JSC.0000000000005000.

lowering the weight under control toward the ground, and then standing up again. You can imagine the stretch to your hamstrings and lower back that movement provides if you use a full range of motion. Including some movements that are most challenging for a given muscle group when it is in the elongated position can help build flexibility and can help you remain "strong in any position," which assists mobility too.[131] Again, if any of this is confusing, I provide a gallery of movements to choose from at protocolsbook.com/videos.

I also hope you'll include the direct mobility exercises that I describe below in your weekly routine but even if you don't, by training the way I describe above, most people will improve their mobility, and flexibility too. We're talking about the ability to touch your toes while keeping your legs straight or just slightly bent, to hang from a bar with no pain, to twist around to look behind you while seated in a chair, to sit in a deep squat for one to two minutes (and be able to stand up at the end!), and so on. You can develop those basic abilities through resistance training with a full range of motion, and focusing on the relevant muscles at all positions in each rep.

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[131] Milo Wolf et al., "Lengthened Partial Repetitions Elicit Similar Muscular Adaptations as Full Range of Motion Repetitions During Resistance Training in Trained Individuals," PeerJ 13 (February 2025): e18904, https://doi.org/10.7717/peerj.18904; Alizadeh et al., "Resistance Training Induces Improvements."

There are many great ways to improve mobility. Here I have created what I believe is the minimal effective dose of it. And if you want to do more mobility work, please do!

At least two to four times a week, take ten minutes and do the following four movements. You can do these most any place and at any time.

### 1 Deep sumo squat, torso twist, point, and look up

Set your feet a few inches wider than your shoulders, then squat down as far as you can.

Put your elbows against the insides of your knees and your palms and fingers together in “prayer” position, and push out gently with your elbows until you feel a stretch in your hips and inner thighs. Then raise one arm and, with your index finger, point directly up at the ceiling (or as high overhead as you can) while keeping your other elbow against the inside of your knee. Let your gaze follow your finger, which will naturally have you twist your torso and turn your neck to raise that arm. Look directly up at your index finger for three to ten seconds. Inhale and exhale.

Then put your arm down, reset to the first squat position with both elbows against the insides of your knees. Place your hands in prayer position for three to ten seconds, and then raise your other arm up to point at the ceiling. Twist your

torso and look up at your index finger for three to ten seconds. Repeat three times for each side. You may have to stand up between sets. That is fine.

This is a great movement because it gets you squatting deep, it improves your thoracic spine and neck mobility, and you'll also feel a decent stretch in your inner thighs and hips. It will also reveal if you have any asymmetries in your mobility (perhaps one side versus the other) which is common.

I demonstrate this movement here: protocolsbook.com/videos.

Note: If you can't point and/or look all the way up to directly overhead, start by pointing and looking up at a lower angle. Gradually progress to pointing and looking directly overhead.

### 2 Get your knees behind your glutes, one at a time

This is one I learned from mobility expert Kelly Starrett whom I mentioned earlier in this chapter. Since it may not be easy for most people to envision, try this: Start in a standing lunge by placing one foot in front of you and the other foot behind you. Bend your front knee and keep your rear leg straight or a bit bent. Now nudge the foot in front of you slightly farther out. Bend that front knee more and focus on stretching the hip flexors at the top of your back leg. You

may need to lean your torso back a bit to really feel the hip flexor stretch. Okay, notice the knee of your back leg is now behind your glutes? That is what I mean by “get your knee behind your glutes.” Slowly ease into this hip flexor stretch for ten to thirty seconds while exhaling. It’s fine to reach up and look overhead while doing this, as that will help extend the hip flexor, but keep your attention on the back leg. Your front leg is there for stability and you can bend it to provide more rear-placed-leg hip stretch. Repeat for each side three times. I demonstrate this movement here: protocolsbook.com/videos.

Next, you’re going to kneel on both knees. Then put one knee up in front of you (the foot of that same leg is thus flat on the ground). The shin of your rear leg is parallel to the ground. That means your back foot will be sole-facing-up. Place your hand on that raised front knee for stability and gently extend the back leg until you feel a stretch in your hip flexor. Again, think “I need to get my knee behind my glutes.” Keep your torso upright, your eyes forward or up, and even lean slightly back at the waist. Hold the stretch for ten to thirty seconds. Repeat three to four times. Don’t forget to breathe.

I demonstrate these movements here: protocolsbook.com/videos.

The reason to do this: Your hip flexors are located where your leg meets your pelvis. When you sit, your hip flexors are partially flexing. When you run, bike, or walk up a set of stairs, your hip flexors are repeatedly flexing as you bring your knee up and forward. Modern life has us flexing our hip flexors all the time. A key step toward good mobility is to make sure you can extend your hips. Hip extension is the opposite of hip flexion—it moves your knee away from the front of your pelvis, to a location behind you. Hence the directive here. Once you experience how good it feels during (and after!) this stretch, you'll be looking for ways to do it more often.

### 3 Get into a reverse tabletop position

Sit on the floor with your feet in front of you, the soles of your feet and butt on the ground, and with your knees bent. Place your hands at your sides a few inches away from your hips, with your palms down. Turn your hands so your fingers point away from you out to each side.

See if you can slowly push your pelvis up so that the tops of your legs and torso are parallel to the floor, and your knees are at a 90 degree right angle. Many people find they can't get into this "reverse tabletop" position for very long, if at all, but some can. See if you can hold that position for ten seconds.

If you can't do this, practice doing this each week for three sets of however long you can hold it until you can easily remain in this position for ten seconds. Once you can do that, move to the following progression:

Sit on the floor with your feet in front of you, with the soles of your feet and butt on the ground, and your knees bent. Place your hands just behind your hips, with your palms down. Turn your hands so your fingers point backward. See if you can slowly push your pelvis up so that the tops of your legs and torso are parallel to the floor, and your knees are at a 90 degree right angle.

Once again, the goal is to get into the flat reverse tabletop position and hold for ten seconds. For some people this will take a while to achieve. For others it will be easy depending on your level of shoulder mobility. These movements are also great for wrist mobility, too, by the way.

Once that final "reverse tabletop" variation is achievable, then you can start to experiment SLOWLY with this next movement. Sit on the floor with your legs straight in front of you, heels on the ground and the soles of your feet facing forward. Place your hands flat on the ground just behind you, with your fingers pointed backward. Keep them there as you gently and slowly scoot your glutes forward. Remember, your hands stay flat on the ground and fixed in place. Keep scooting forward until you feel a stretch in your biceps and

shoulders. Exhale, and see if you can ease into it another centimeter or so. Once you can extend your shoulders back behind you with the edges of your hands touching, you are getting to quite mobile shoulder status.

The reason to do this: Tight shoulders can arise from many sources, but often there will be people who can raise their arms to the front or to the side without issue but who don't have the mobility to raise their arms up behind them, meaning they don't have much shoulder extension mobility. This is made worse by desk work, and the chin-down/forward-leaning posture that is nearly ubiquitous nowadays, thanks to excessive amounts of time texting and scrolling on our phones. Shoulder tightness and neck and back pain are often linked too. Improving your shoulder extension mobility is straightforward. It just requires a willingness to work through these progressions.

Wrist, back, knee, hip, and shoulder injuries are super common and many can be avoided by a quality strength training program and by including some basic mobility exercises like the ones in this protocol. Some quick movements you can incorporate in your routine are the "face pulls," which strengthen these and other areas, as well as neck training a few times a week, as described in Exercise Protocol 2 and shown at protocolsbook.com/videos. You

will be amazed at how strong, mobile, and stable your shoulders, neck, and upper back will be if you do.

Do not push too fast when it comes to mobility work! Work through the progressions listed above over the course of months or more. Make it painless but deliberate, and you will build and maintain much greater shoulder, hip, knee, back, and wrist mobility.

### 4 Practice the Founder pose

The final mobility exercise is the Founder pose, made popular by chiropractor Eric Goodman, who has served as a conditioning coach for numerous top athletes in sports ranging from cycling to surfing to water polo. I was referred to his work by friends who previously had back pain and shoulder pain, and who sit a lot of the day. “You have to try this thing called Founder pose,” I kept hearing. Mind you, in my role, I hear that about a thousand times per day about a variety of things, so I’m not easily moved to try them. But I heard this decompresses the spine and trains your musculature to work properly as a functional unit.

I tried Eric’s Founder pose and, wow, it’s great. Not just because it feels good while you do it but your posture improves. And it definitely made me more flexible.

So what is it? It’s similar to the “chair pose” in yoga but while that mainly emphasizes the musculature of the front of

the body, the Founder pose emphasizes mobility of your posterior and lateral chains: your hamstrings, glutes, lower and upper back, and thoracic muscles.

Here is how you do it: Stand with your feet a bit wider than your shoulders, toes pointing slightly out.

Stand as tall as you can by pressing your heels down and lifting your head up, as if a string were tied to the top of your head but your heels were fastened down to the ground.

Hinge your hips so you lean slightly forward at the waist—almost as though you're about to sit in a chair but freeze only about a third of the way down into the movement. Your hips and glutes are back behind your heels, and your knees are just slightly bent. Keep your head as far away from your glutes as possible. You should feel a stretch in your lower back and hamstrings. If not, lean farther forward at the waist until you do.

Raise your chest, and with your arms at your sides, turn your thumbs out and away from the midline of your body (away from your belly button).

Keeping your thumbs rotated out, start to slowly lift your arms. Stop when you feel more stretching in your lower back and/or the backs of your legs.

Now move your hips back a few inches more. And raise your chest and head, aka “make your spine long,” still pressing your heels down.

Raise your arms a bit more if you can (thumbs still pointed away from your navel in external rotation), all the while pushing your heels down.

Hold this position for five more seconds then stand up straight.

Repeat that one to two times, as often as every day or even just a few times a week.

Thank me later. No, thank Eric Goodman!

I demonstrate this movement here: protocolsbook.com/videos.